



Undergraduate / Systems and Computer Engineering
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Privacy Protection in Location Data through Differential Privacy: Evaluation of Disturbance Mechanisms.

by

Carol Sofia Florido Castro

Advisors:

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Val'erie Gauthier Umana, Juan Fernando Perez Bernal

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Chapter 1

Introduction

The proliferation of mobile applications and location-based services has transformed radically changed the way personal data is collected, processed, and used. Since mobile devices became inseparable extensions of life In everyday life, the continuous collection of location information has become not only commonplace, but expected [1]. GPS navigators, mobility platforms, social networks, applications of Appointments, delivery services, and proximity-based advertising systems depend on access constant reliance on users' location data. This dependence, however, has made location one of the most vulnerable vectors for the involuntary exposure of personal information.

Unlike other types of data, location records allow us to infer aspects deeply sensitive and structural aspects of an individual's life: their daily habits, patterns of movement, address, workplace, religious practices, state of health and even their social or political ties. Recent studies have shown that a small... number of geographical points to re-identify a person with a high probability, even if The data has been anonymized. For example, it has been verified that with just four points Randomly selected spatiotemporal individuals can uniquely identify 95% of the individuals in a dataset of 1.5 million people, demonstrating uniqueness. of human mobility patterns [2]. This has raised increasing concerns in the academic community, regulatory bodies, companies, and among the users themselves, who, while desiring personalized and contextually relevant services, also demand greater control over your privacy.

This scenario is framed within a broader context of tensions between technological innovation Logic and fundamental rights. As regulations such as the Regulation become more firmly established General Data Protection Regulation (GDPR) in Europe, the Consumer Privacy Act of With California (CCPA) and other emerging legal frameworks in Latin America and Asia, organizations face increasing pressure to implement techniques that allow them to exploit value in the data without compromising individual privacy [3]. At this point, differential privacy has emerged as one of the most robust and formally guaranteed paradigms for protect personal data [4].

Differential privacy, by introducing statistically controlled noise onto the data or the answers to queries, allows for useful analyses without revealing specific information about any individual. This technique ensures that the aggregate results reflect trends and patterns of behavior without compromising the identity or data of a specific person, establishing a balance between analytical utility and privacy protection. Its adoption by leading companies such as **Apple** [5] and **Google** [6] has reinforced its technical and operational legitimacy, showing practical applications in real data environments. Apple uses differential privacy to improve Apple Intelligence products and services, such as Genmoji, image generation tools, and email text summaries. The company applies controlled noise and aggregation techniques to collect only aggregated signals from the devices of users who have opted into its Device Analytics program. This allows Apple to identify usage trends and evaluate effectiveness of their AI models and optimize functionalities, without accessing individual data or personal content, ensuring that sensitive information, such as specific prompts or emails, never link to a specific user. For its part, Google (BigQuery) enforces privacy differential in large dataset analysis queries, limiting the contribution of each entity to a group. It groups each entity into multiple groups, restricting aggregate values and eliminating noise from the final results. It also removes groups with few entities to avoid re-identification. These measures allow companies and analysts to gain significant insights into behaviors and trends of user groups without exposing individual information, which is crucial in contexts of risk of re-identification and to comply with regulations of privacy. Taken together, these applications show that differential privacy is not only not a theoretical standard, but a strategic and operational tool for companies seeking to maximize the value of their data while protecting the privacy of their users, strengthening trust and compliance with regulatory frameworks such as GDPR and CCPA.

At the institutional level, the United States adopted differential privacy to protect data of the 2020 Census, replacing traditional methods such as record swapping. This allowed for guaranteeing the confidentiality of citizens even at the level of census blocks, although it introduced some loss of precision in very small geographical units, especially among minority groups. This case demonstrates how differential privacy can apply to real and critical data, balancing protection and utility [7].

In particular, protecting user journeys under the differential privacy model presents unique technical challenges. The trajectories exhibit strong spatiotemporal dependence, high dimensionality, and structural complexity. Furthermore, the users have different disturbance tolerance levels, depending on the context of use, the type of application and their own perceptions of risk. Therefore, it is necessary to carefully evaluate the available mechanisms, understanding how they affect not only guaranteed privacy, but also the practical utility of the data for different types of services.

This project proposes to compare different differential privacy methods applied to location data, in order to evaluate how they affect the analyses performed based on the trajectories obtained and how well they adapt to different user profiles.

integrate quantitative criteria such as expected distance, loss of spatial precision and the preservation of mobility patterns, along with customizable privacy parameters,

This thesis seeks to contribute to the design of systems that protect location data in a flexible, robust, and user-centered. To facilitate this evaluation, a

An interactive tool that allows you to visualize GPS trajectories and apply different algorithms.

of differential privacy and graphically observe the impact of the mechanisms on

mobility patterns. The analysis will be performed using **Origin-Destination matrices** to

characterize mobility flows and assess how spatial and spatial patterns are preserved

displacement. Likewise, the behavior of the algorithms will be explored under different

Scenarios will be analyzed and their performance will be compared graphically, allowing for the identification

of advantages, limitations, and the degree of distortion introduced by each mechanism. Finally, the aim is

To offer guidelines that guide the practical implementation of differential privacy mechanisms in mobility,

geolocation and urban analysis systems, ensuring a balance

between innovation, utility and protection of individual rights.

Chapter 2

Differential Privacy

Differential Privacy (DP) is a mathematical framework designed to protect individual information within a dataset, ensuring that the result of any statistical analysis is not significantly affected by the inclusion or exclusion of a single individual. In Dwork's words [8], a function satisfies privacy differential if the probability of obtaining a certain result is almost the same regardless of whether an individual record is present in the database. Formally, a randomized mechanism K satisfies differential ϵ -privacy if, for every pair of bases of data D and D' that differ in a single row, and for any subset of possible outputs S , it follows that

$$Pr[K(D) \in S] \leq e^{\epsilon} \times Pr[K(D') \in S],$$

where the parameter ϵ (ϵ) quantifies the level of privacy, controlling the maximum factor by which the probabilities of obtaining a result when modifying a single data point in the set. Small values ($\epsilon < 1$) guarantee greater privacy when making the produce almost indistinguishable outputs, while larger values ($\epsilon > 5$) allow for greater precision but They reduce the protection. That is, ϵ determines how much a perturbed piece of data can reveal about the user's actual position. In the context of location data, this principle can be applied to protect the privacy of users who share their geographic coordinates with services that rely on location-based information, such as navigation applications, nearby recommendations or urban mobility analysis. Based on this, it is considered that User location data is protected by a randomized K mechanism which guarantees differential privacy. In this way, any analysis performed on the The released data does not allow the exact location of any individual to be inferred, preserving their sensitive information. For example, places visited, associated semantic tags (hospital, church or protest) and activities inferred from the location are protected.

Figure 2.1 illustrates the flow of conceptual information of the differential privacy mechanism. rencial applied to location coordinates.

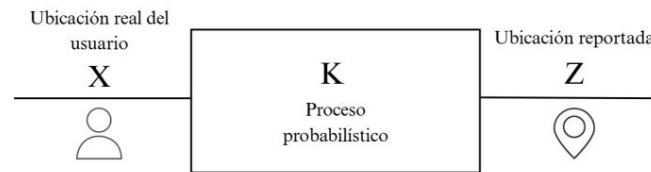


Figure 2.1: Information flow of the differential privacy mechanism applied to location coordinates.

Formally, K can be defined as a randomized function:

$$K : X \rightarrow P(Z)$$

where X represents the set of real locations (e.g., coordinates (lat, lon)) and $P(Z)$ is the set of probability distributions over the released coordinates Z .

The mechanism produces an output $z \in Z$ that is used in analysis or services, balancing privacy and utility without assuming that each individual user applies the disturbance. From the perspective of quantitative information flow, K can be modeled as a channel with input x and output z , where the output can include spatial generalizations, for example, moving from exact coordinates to a zone, neighborhood, or city.

2.1. Differential Privacy Models: Central Differential Privacy (CDP) and Local Differential Privacy (LDP)

The literature primarily distinguishes two approaches to applying differential privacy: central differential privacy (CDP) and local differential privacy (LDP) [9]. In the central differential privacy (CDP) model, a trusted database administrator with direct access to individual data is assumed. This administrator applies a randomization mechanism K that guarantees differential ϵ -privacy before releasing any aggregated results or statistics. This model is suitable when a trusted entity exists that can safeguard sensitive information and apply the necessary perturbations to preserve privacy. However, in many scenarios, a trusted central entity does not exist, or the goal is to avoid aggregating sensitive data into a single repository. In these cases, the local differential privacy (LDP) model, also known as the fully distributed model, is adopted. In this approach, each user maintains control over their own data and applies a local randomization mechanism, called the local randomizer R , which transforms their information before sharing it. Formally, a local ϵ -randomizer $R : X \rightarrow W$ is an algorithm that satisfies differential privacy on a single-record basis [9, Section 12.1]. In the size-one data model, i.e., on an LDP, the interactions between the algorithm and the data are performed through a local randomization oracle $\text{LRD}(x, R)$, which takes as input an index i and a local randomizer R , and returns $R(x, x)$. If the algorithm

a response w generated according to the distribution

performs multiple invocations to the oracle on the same individual with different randomizers

$\{R_1, R_2, \dots, R_k\}$, the total privacy of the process is obtained through **additive composition of the privacy parameter**

$$\epsilon_{\text{total}} = \epsilon_1 + \epsilon_2 + \dots + \epsilon_k.$$

This property, known as the sequential composition theorem [10], reflects that each additional interaction with the data implies a cumulative loss of privacy. In the context of local differential privacy (LDP), where each user applies independently the randomization mechanism on its own data, this loss becomes especially relevant. Every time a user posts a new perturbed coordinate or a new Trajectory tracking uses a portion of the total privacy budget, progressively increasing the risk of inferring information about the actual location. For this reason, in scenarios where multiple points are perturbed, it is necessary to consider values of ϵ larger than compensate for the number of perturbations performed. In other words, as the number of perturbed coordinates increases, the local differential privacy (LDP) model requires a larger global ϵ to maintain an acceptable level of utility in the released data.

Dwork and Roth [9, Section 12.1] point out that, although local differential privacy (LDP) It offers greater guarantees by eliminating the need for a reliable central entity, but suffers from a faster loss of utility due to this additive composition. In the context of trajectory perturbation, this implies that the design of the mechanism must carefully balance the level of injected noise with the number of published points, so that a reasonable balance between privacy and spatial accuracy is maintained.

2.2. Overview of the thesis content

This thesis addresses the protection of privacy in location data through the Differential Privacy (DP) framework, exploring both centralized (CDP) and Local (LDP). Probabilistic coordinate perturbation mechanisms are studied, including the Laplace Mechanism, Pivot Sampling and Anchor-Based Pivot Sampling, evaluating Its performance in terms of the balance between privacy and utility is analyzed in detail. the role of the privacy budget ϵ , highlighting how its configuration affects the magnitude from the noise introduced and the loss of information in user trajectories. Throughout The thesis presents experiments on real trajectories in Beijing, where they are constructed Source-Destination matrices to quantify flows between cells and serve as the basis for analysis graphic. These matrices allow us to evaluate the spatial distortion introduced by the mechanisms of disturbance and facilitate comparison between algorithms. In addition, interactive graphical visualizations are developed that allow exploration and evaluation of the impact of the disturbance. in spatial data, illustrating the relationship between accuracy, privacy, and configuration of experimental parameters. Finally, visual tools are provided to help to understand the behavior of the different differential privacy mechanisms.

2.3. Laplace Mechanism for Coordinate Perturbation

The Laplace mechanism for privacy protection in coordinates [11] falls within the **central differential privacy** (CDP) model and is based on the controlled addition of noise following a Laplace distribution. Given an original point x a privacy parameter $\tilde{y} > 0$, the process generates a perturbed point z through a probabilistic transformation. ² and The implemented perturbation mechanism follows the approach proposed in [11]. The probability density for the perturbed point z is given by

$$D\tilde{y}(z|x) = \frac{2\tilde{y}}{2\tilde{y}} e^{-\tilde{y}d(x,z)},$$

where $d(x, z) = \sqrt{(x_1 - z_1)^2 + (x_2 - z_2)^2}$ is the Euclidean distance between points x and z .

The term $\frac{2\tilde{y}}{2\tilde{y}}$ It acts as a normalization factor, ensuring that the probability density function is valid and that the total probability integrates to 1. This formulation guarantees that points closer to the original have a higher probability of being selected (due to $\tilde{y}d(x,z)$), while those further away have a decreasing probability.

at the end and

exponentially with distance.

Based on the above, to efficiently implement this mechanism in the context of location data, a representation in x -centered polar coordinates must be used.

A point z is expressed as $(r, \tilde{y})$, where r is the radial distance from x and $\tilde{y}$ is the polar angle.

The joint density at these coordinates factors as

$$D\tilde{y}(r, \tilde{y}) = \underbrace{\frac{1}{2\tilde{y}}}_{\text{uniform in angle}} \underbrace{2r e^{-\tilde{y}r}}_{\text{radial distribution}}.$$

The generation of the perturbed point takes advantage of this factorization. First, an angle $\tilde{y}$ is selected uniformly in $[0, 2\tilde{y})$. For the radial component, the cumulative distribution function is used.

$$C\tilde{y}(r) = 1 - \tilde{y} (1 + \tilde{y}r) e^{-\tilde{y}r}.$$

Using the inverse transform method, a uniform value $p \in [0, 1)$ is generated and $C\tilde{y}(r) = p$ is solved to obtain the radius

$$r = \tilde{y} \left(\frac{1}{\tilde{y}} - W_{-1} \left(\frac{p - 1}{\tilde{y}} \right) \right),$$

where W_{-1} is the negative branch of the Lambert function W , defined as the inverse function of $f(W) = W e^W$ for $W \leq -1$. That is, for any $y \in [-1/e, 0)$ it holds that $W_{-1}(y) e^{W_{-1}(y)} = y$, with $W_{-1}(y) \leq -1$. Finally, the Cartesian coordinates of the perturbed point are calculated as

$$z = x + (r \cos \tilde{y}, r \sin \tilde{y}).$$

This mechanism preserves location privacy, ensuring that the generated points do not definitively reveal the original position. In particular, the points closest to the position

The original points x have a higher probability of being selected, while more distant points appear with a probability that decreases exponentially with distance. The parameter $\tilde{y}$ controls the level of noise introduced: larger values make the disturbed points closer to the original (less protection), while smaller values produce greater dispersion (more privacy).

Illustrative example. Consider an original point $x = (0, 0)$ and a privacy budget $\tilde{y} = 1$. To generate a perturbed point, first obtain an angle $\tilde{y}$ sampled uniformly over the interval $(0, 2\tilde{y})$; in this case, assume that $\tilde{y} = \tilde{y}/4$ (45°). Then generate a value $p \sim \text{Uniform}(0, 1)$, obtaining $p = 0.5$. From this value, solve the equation $1 - \tilde{y} (1 + r)e^{-\tilde{y}r} = 0.5$, which results in $r \approx 1.67835$. Finally, calculate the perturbed Cartesian coordinates as

$$z = (0 + 1.67835 \cos(\tilde{y}/4), 0 + 1.67835 \sin(\tilde{y}/4)) \approx (1.1868, 1.1868).$$

This result shows how the original point $(0, 0)$ is displaced $(1.1868, 1.1868)$ by the perturbation mechanism, maintaining differential privacy.

Conversion of geographic coordinates to a local system in meters

GPS coordinates are expressed in degrees of latitude and longitude. However, these units are not uniform in terms of distance: one degree of latitude corresponds approximately to 111 km, while one degree of longitude varies with latitude, being greatest at the equator and decreasing towards the poles. For this reason, a small numerical disturbance in degrees can translate into a very large displacement in meters, which complicates the control of the magnitude of the noise introduced by the Laplace mechanism.

For the parameter $\tilde{y}$ to have a clear physical meaning, such as “introducing a mean noise of 20 m”, a projection to a local, planar coordinate system, measured in meters, is first performed. For paths covering relatively small areas (such as routes within a city), the equirectangular projection can be used, which approximates the Earth's surface as a tangent plane at a reference point $(\text{lat}_0, \text{lon}_0)$, chosen here as the first point of the path. The conversion from latitude-longitude to planar coordinates (x, y) in meters is performed as follows:

$$x = (\tilde{y} - \tilde{y}_0) \times \cos(\tilde{y}_0) \times R, \quad y = (\tilde{y} - \tilde{y}_0) \times R,$$

where $\tilde{y}$ and $\tilde{y}$ are the longitude and latitude (in radians), $\tilde{y}_0$ and $\tilde{y}_0$ are the coordinates of the reference point, and $R \approx 6,371,000$ m is the mean radius of the Earth. Once the perturbation in meters has been applied using the Laplace mechanism, it is necessary to return to geographic coordinates for visualization on a map. To do this, the previous formulas are inverted.

$$\tilde{y} = \frac{x}{R \cos(\tilde{y}_0)} + \tilde{y}_0, \quad \tilde{y} = \frac{y}{R} + \tilde{y}_0,$$

$\check{y}$ and $\check{y}$ are then converted to degrees. With the trajectory expressed in local coordinates (x, y) , the Laplace mechanism is applied. In this system, the expected magnitude of the radial displacement $E[r]$ is related to the privacy parameter $\check{y}m$ (expressed in m°)

as

$$E[r] = \frac{2}{\check{y}m}.$$

For example, if an average displacement of 20 m is desired, it is calculated

$$\check{y}m = \frac{2}{0.1 \text{ m } 20^\circ} = 0.1 \text{ m } 20^\circ.$$

To understand how the above formula is derived, it is necessary to consider that in polar coordinates $(r, \check{y})$ centered on the real point, the joint density of the radius and the angle is expressed

as

$$f_{R,\check{Y}}(r, \check{y}) = \frac{\check{y}_m^2}{2\check{y}} r e^{\check{y}mr},$$

where $r \geq 0$ and $\check{y} \in [0, 2\check{y})$. Integrating over $\check{y}$ gives the marginal density of the radius

$$f_R(r) = \check{y} \frac{2}{m} r e^{\check{y}mr}.$$

The expected magnitude of the radial displacement is then calculated as

$$E[r] = \int_0^{\check{y}} r f_R(r) dr = \check{y} \frac{2}{m} \int_0^{\check{y}} 2 r e^{\check{y}mr} dr.$$

Using the general formula

$$\int_0^{\check{y}} r^n e^{-ar} dr = \frac{n!}{n+1, a} \quad a > 0,$$

and considering $n = 2$ and $a = \check{y}m$, we have

$$\int_0^{\check{y}} 2 r e^{\check{y}mr} dr = \frac{2}{\check{y}_m^3}.$$

Substituting into the expression for $E[r]$

$$E[r] = \check{y} \frac{2}{m} \times \frac{2}{\check{y}_m^3} = \frac{2}{\check{y}m}.$$

Therefore, the expected magnitude of the radial displacement is inversely related to the privacy parameter.

$$E[r] = \frac{2}{\check{y}m}.$$

2.3.1. Analysis of the perturbation graph

Figure 2.2 shows the effect of the Laplace mechanism applied to a diagonal path of $N = 20$ points with a privacy budget $\check{y} = 1$. It can be seen that the perturbed points are scattered around the original path, reflecting the random nature

of Laplace noise. Each point moves away from its original position approximately following the expected exponential radial density of the mechanism. Given the relatively low value of $\tilde{y}$, the dispersion is wide, which increases privacy but reduces the accuracy of the trajectory. Despite the addition of noise, the overall trend of the diagonal trajectory is maintained, demonstrating that the mechanism preserves the global shape of the trajectory and only alters the individual location of the points. The larger markers for the initial and final points clearly identify the orientation of the trajectory, showing that the mechanism respects the temporal sequence of the points and does not generate reversals in the order of the trajectory. The value of the parameter $\tilde{y}$ directly influences the dispersion of the points: with $\tilde{y} = 1$, the expected average displacement is considerable, as can be seen in the figure. Increasing $\tilde{y}$ would reduce dispersion, bringing disturbed points closer to their original positions and decreasing differential privacy, while lower values of $\tilde{y}$ would increase dispersion, increasing the protection of individual data.

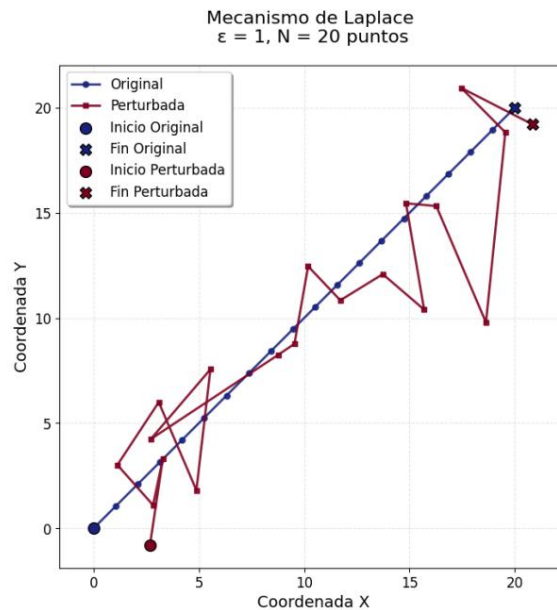


Figure 2.2: Original and perturbed trajectory with $\tilde{y} = 1$ and $N = 20$ points. The circles and crosses indicate the start and end points of each trajectory, respectively.

In summary, the graph demonstrates how the Laplace mechanism introduces noise in a controlled manner, preserving the privacy of each individual coordinate while maintaining the overall consistency of the trajectory. This visualization allows for an intuitive understanding of the balance between precision and privacy controlled by the parameter $\tilde{y}$.

2.4. Pivot Sampling Algorithm for Path Privacy (TP)

The Pivot Sampling algorithm is a **local differential privacy** (LDP) mechanism that It protects geographical trajectories through perturbation in two stages: independent perturbation of pivot points and directional perturbation of non-pivot points. This explanation is based on what was proposed by [12].

2.4.1. Fundamental Concepts

Given an original path $\tilde{y} = p_i \tilde{y} P \mid 1 \tilde{y} i \tilde{y} |\tilde{y}|$, it is defined as an ordered sequence of points p_i belonging to a spatial domain P , which represents the set of all possible locations. Each point p_i is composed of a pair of coordinates (x_i, y_i) that indicate its position in two-dimensional space. The length of the path, denoted by $|\tilde{y}|$, corresponds to the total number of points that make it up. First, the algorithm creates two identical copies $\tilde{y}'$ and $\tilde{y}''$ of $\tilde{y}$. In each copy, a subset of points is selected selected alternately known as **pivot points**, (e.g. even or odd index points) that are perturbed independently using the Exponential Mechanism (EM)

$$P(r|\hat{p} = r) = \frac{\exp \left(\frac{\tilde{y}ind \times u(p, r)}{2\tilde{y}u} \right)}{\sum_{r' \in P} \exp \left(\frac{\tilde{y}ind \times u(p, r')}{2\tilde{y}u} \right)}. \quad (2.1)$$

In the context of trajectory perturbation, an original real point p belonging to the trajectory and its corresponding perturbed version $\hat{p}$ are considered. Both points are part of the spatial domain P , which contains all possible locations. Within this domain, r represent two candidate points that could be selected and r' as perturbed versions.

bases of the original point. To decide between these candidates, a utility function defined as $u(p, r) = \tilde{y}dist(p, r)$ is used, where $dist(p, r)$ represents the Haversine distance between the points $p = (lat1, lon1)$ and $r = (lat2, lon2)$. This distance is calculated using the formula

$$dist(p, r) = 2R \arcsin \left(\sqrt{\sin^2 \frac{\tilde{y}lat}{2} + \cos(lat1) \cos(lat2) \sin^2 \frac{\tilde{y}lon}{2}} \right), \quad (2.2)$$

where R is the radius of the Earth (approximately 6,371 km), $\tilde{y}lat = lat2 - lat1$ and $\tilde{y}lon = lon2 - lon1$, expressed in radians. The negative sign in the utility function allows the exponential mechanism to favor the selection of nearby points, since these offer greater utility (shorter distance).

The sensitivity of function (2.1), denoted as $\tilde{y}u$, is defined as the maximum possible change in utility when comparing two different candidates with respect to the same point, that is, $\tilde{y}u = \max_{\tilde{y}p, r, r'} |u(p, r) - u(p, r')|$. Finally, the parameter $\tilde{y}ind$ represents the privacy budget associated with the perturbation of each point independently, which determines the degree of protection guaranteed against adversarial inference. To select the pivot points that will serve as the basis for the perturbation, the algorithm implements a mechanism of

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strategic exchange that distributes the points of the original trajectory between the two copies: $\tilde{y}$ (that is, $\tilde{y}$). Initially, the even points of the trajectory are assigned as pivots at $\tilde{y}$ is, $P_{\tilde{y}ind} = \{p_2, p_4, \dots\}$, while the odd ones are used as pivots in $\tilde{y}$ (i.e., $P_{\tilde{y}} = \{p_1, p_3, \dots\}$). Note that the relation $P_{ind\tilde{y}} = P_{\tilde{y}} P_{ind\tilde{y}}$ is satisfied, providing total $\tilde{y}$ which ensures a and non-redundant coverage of the set of points. During the perturbation process, even points are on the trajectory $\tilde{y}$ perturbed independently by the Exponential Mechanism, while odd points take advantage of directional information from their already perturbed neighbors. In the trajectory $\tilde{y}$, the reverse process occurs: odd points are perturbed independently, and even points use information from their odd neighbors to guide their perturbation. This exchange of roles promotes a balance between privacy and utility, as it prevents the entire trajectory from depending solely on independent or correlated perturbations.

Illustrative example. Consider a path $\tilde{y} = \{p_1, p_2, p_3, p_4, p_5\}$. Table 2.1 shows how the points are assigned between the two copies of $\tilde{y}$ in the strategic change: $\tilde{y}$ and $\tilde{y}^*$ during the inter- process $\tilde{y}$, the even points of the original path are chosen as **pivots** in $\tilde{y}$ (p_2 and p_4) and the odd points are assigned as **non-pivots** (p_1, p_3, p_5), while in $\tilde{y}^*$ the reverse process occurs, with the odd points as pivots (p_1, p_3, p_5) and the even points as non-pivots (p_2, p_4). This distribution can be visualized directly in the table.

Table 2.1: Point allocation in the exchange process

Copy	Pivot Points	Non-Pivot Points
$\tilde{y}^*$	p_2, p_4	p_1, p_3, p_5
$\tilde{y}$	p_1, p_3, p_5	p_2, p_4

2.4.2. Directional Disturbance

For non-pivot points, the algorithm introduces noise in the direction relative to neighboring pivot points in order to preserve privacy while maintaining some spatial coherence. To do this, the angular space is divided into a finite number of equal sectors, each representing a possible direction. The actual direction of a point is initially assigned to one of these discrete sectors according to its geometric angle. This is done by means of the following formula

$D = \{0, 1, \dots, |D| - 1\}$, where each $d \in D$ covers $(2d + 1) \frac{\tilde{y}}{|D|}$ (23)

Next, the k-RR randomized response mechanism is applied, which introduces uncertainty into the reported direction: the true sector is selected with high probability (controlled by the privacy parameter $\tilde{y}$), but another sector is chosen at random with a small probability. This ensures that the final reported value does not reveal the exact original direction, making precise point identification difficult, but without completely losing directional information, thus helping to preserve the usefulness of the perturbed trajectory. The mechanism

k-RR then operates as

$$Pr[K(d) = d'] = \begin{cases} \frac{\tilde{y}}{|D|\tilde{y} + e} & \text{if } d = d' \quad (\text{correct direction}) \\ \frac{1}{|D|\tilde{y} + e} & \text{if } d \neq d' \quad (\text{incorrect direction}), \end{cases} \quad (2.4)$$

where:

- K : Disturbance mechanism
- $\tilde{y}$: Privacy budget for addresses
- d : Original address (sector index)
- d' : Reported address

Illustrative example. Suppose the angular space is divided into $|D| = 8$ sectors. Each radian (45 degrees). If sector covers an angle of $\tilde{y} = \frac{2\tilde{y}}{8} = \frac{\tilde{y}}{4}$ a point has a relative direction $\frac{5\tilde{y}}{8}$, then this angle falls within the sector

$$(2d - 1)\frac{\tilde{y}}{|D|(2d + 1)} \leq \frac{\tilde{y}}{|D|} \leq (2d + 1)\frac{\tilde{y}}{|D|(2d + 1)}$$

Finding the value of d that satisfies this inequality for $\tilde{y} = \frac{5\tilde{y}}{8}$, we obtain $d = 3$, because

$$(2 \times 3 - 1)\frac{\tilde{y}}{8} \leq \frac{5\tilde{y}}{8} \leq (2 \times 3 + 1)\frac{\tilde{y}}{8}$$

Therefore, the original address is assigned to sector $d = 3$. Then, using the k-RR mechanism with $\tilde{y} = 1.0$, the probability of reporting the true address is calculated as

$$Pr[K(3) = 3] = \frac{1}{e} + \frac{2,718}{1 + 2,718} \approx 0.287$$

and the probability of reporting any of the incorrect sectors ($d \neq 3$) as

$$Pr[K(3) = d] = \frac{1}{1 + e} \approx \frac{1}{9.718} \approx 0.103.$$

This example shows how randomness is introduced into the direction without completely eliminating spatial information.

2.4.3. Determination of the Disturbance Domain

each non-pivot point p_j , is defined as the set of possible locations for p_j in space that are restricted to a specific angular range determined by the perturbed directions from its neighboring pivot points. This domain is calculated using a procedure that, given a perturbed pivot $\hat{p}^k$ and a perturbed angular direction $\hat{d}_{k,j}$, obtains all points $p \in P$ whose relative direction with respect to $\hat{p}^k$ falls within the angular sector bounded by the angles $\tilde{y}_{\min}$ and $\tilde{y}_{\max}$.

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limits are determined according to the angular sector index $\hat{d}_{k,j}$ and the total number of sectors $|D|$, to the so that

$$\hat{y}_{\min} = (2 \hat{d}_{k,j} - 1) \frac{\bar{y}}{|D|}, \quad \hat{y}_{\max} = (2 \hat{d}_{k,j} + 1) \frac{\bar{y}}{|D|}.$$

The final perturbation domain for point p_j results from the intersection of the angular sectors defined by the neighboring pivots p_{j-1} and p_{j+1} , that is,

$$P_j = S(p_{j-1}, \hat{d}_{j-1,j}) \cap S(p_{j+1}, \hat{d}_{j+1,j}),$$

where $S(\hat{p}_i, \hat{d}_{i,j}) = \{p \in P : \angle(\hat{p}_i, p) \in I(\hat{d}_{i,j})\}$ represents the set of points whose angle with respect to the origin $\hat{p}_i$ belongs to the angular interval $I(\hat{d}_{i,j})$ associated with the discrete direction $\hat{d}_{i,j}$. This intersection restricts the possible space of the non-pivot point, preserving directional coherence and limiting perturbation to maintain the usefulness of the trajectory.

Illustrative example. Suppose that the space P is discretized into a grid and that there are 8 directional sectors (i.e., $|D| = 8$), each of which assigns a perturbed direction $\hat{d}_{j-1,j} = \frac{\bar{y}}{8}$ radians. For the point 3 to its previous neighbor p_{j-1} , while it has not pivoted p_j , and which assigns $\hat{d}_{j+1,j} =$ subsequent neighbor p_{j+1} . Then

$$\text{From } p_{j-1} : \hat{y}_{\min}^{(1)} = \frac{3\bar{y}}{8}, \quad \hat{y}_{\max}^{(1)} = \frac{5\bar{y}}{8}$$

$$\text{From } p_{j+1} : \hat{y}_{\min}^{(2)} = \frac{5\bar{y}}{8}, \quad \hat{y}_{\max}^{(2)} = \frac{7\bar{y}}{8}$$

Therefore, the allowed angular domain for p_j is contained within a minimal $\frac{5\bar{y}}{8}$ and $\frac{5\bar{y}}{8}$, that is, a intersection (a single shared edge), which significantly restricts its possible location. This type of intersection guarantees that the direction between p_j and its neighboring pivots does not change abruptly, maintaining the spatial consistency of the original path.

2.4.4. Selection of Optimal Granularity

The selection of the optimal granularity, denoted as $|D|$, is fundamental to achieving an adequate balance between precision and privacy in directional disturbance.

This process is formalized as the search for the granularity g within a set of candidates D_c that maximizes the objective function defined as the weighted average of the geometric coverage and the probability of success over different angular ranges of consultation.

Mathematically, this is expressed as

$$|D|_{\text{opt}} = \arg \max_{g \in D_c} \frac{1}{|R(g)|} \left(\underbrace{\bar{y}(d_i; \bar{y})}_{\text{Geometric cover}} \times \underbrace{\bar{y}(d_i; \bar{y}; g)}_{\text{Probability of success}} \right).$$

The reference sets involved in this calculation are: the set of candidate granularities $D_c = \{g_1, g_2, \dots, g_n\}$, which represent the possible numbers of angular sectors; the set of direction indices for a given granularity, $R(g) = \{0, 1, \dots, g-1\}$;

and the set of query angular ranges $\tilde{y} = \{\tilde{y}/2, \tilde{y}/4, \tilde{y}/6, \tilde{y}/12\}$, which define different angular amplitudes of interest. Next, the geometric coverage and probability of success functions are explained.

The geometric cover function $\tilde{y}(di; \tilde{y})$ quantifies what fraction of the angular sector di is contained within the query range $\tilde{y}$, calculated as the ratio between the intersection area of the sector with the query range and the total area of the sector

$$\tilde{y}(di; \tilde{y}) = 2\tilde{y}/g \frac{[(2di-1)\tilde{y}, (2di+1)\tilde{y}] \cap [\tilde{y}/2, \tilde{y}]}{g}.$$

Illustrative example. For example, for a granularity $g = 4$ and a query range $\tilde{y} = \tilde{y}/4$, the geometric coverage for sector $di = 0$ is $\tilde{y}(0; \tilde{y}/4) = 0.5$, indicating that only half of the sector is within the range of interest.

The function $\tilde{y}(di; \tilde{y}; g)$ models the preservation or success probability of the k-RR perturbation mechanism, which depends critically on the privacy budget $\tilde{y}$. This function assigns a higher probability to the correct direction and a lower probability to the incorrect ones according to

$$\tilde{y}(di; \tilde{y}; g) = \frac{e^{\tilde{y}}}{\tilde{y} g^{1+e \tilde{y}}} \quad \text{if } di = d \text{ (correct direction)}$$

$$\frac{1}{1 + \tilde{y} g^{1+e \tilde{y}}} \quad \text{if } di \neq d \text{ (wrong direction)}.$$

Thus, a higher value of $\tilde{y}$ increases the probability of selecting the correct direction, improving the mechanism's accuracy. Regarding design criteria, coarse granularity, with a small value of $\tilde{y}g$, offers a higher probability of success, but generates wider perturbation domains, which can affect spatial accuracy. On the other hand, fine granularity, with a large value of $\tilde{y}g$, allows for more precise perturbation domains, although it reduces the probability of preserving the exact direction due to greater dispersion.

the selection of sectors.

2.4.5. Final Composition and Privacy Analysis

The final composition process seeks to optimize the perturbed trajectory by combining the two versions obtained through the even and odd pivot mechanism. Formally, a trajectory $\tilde{y}^*$ is defined that minimizes the sum of the Haversine distances between each candidate point $\tilde{p}^i$ and its corresponding perturbed points in both trajectories $\tilde{y}^e$ and $\tilde{y}^o$

$$\tilde{y}^* = \arg \min_{\tilde{y}} \sum_{i=1}^{|\tilde{y}|} dHav(\tilde{p}^i, \tilde{p}^e) + dHav(\tilde{p}^i, \tilde{p}^o).$$

Here, $\tilde{y}^e$ represents the perturbed trajectory where the even points act as pivots, $\tilde{y}^o$ corresponds to the trajectory with pivots at odd positions, while $\tilde{y}^*$ corresponds to the trajectory with pivots at odd positions. The goal is to balance the influence of both disturbances to obtain a final representation that reduces the accumulated error compared to the individual trajectories. The optimization process is performed for each point i of the original trajectory, generating a set of

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of candidates P_{cand} within the restricted domain defined by the intersection of neighboring domains. For each candidate, the sum of the Haversine distances with respect to the corresponding perturbed points in $\tilde{y}$ is calculated, $\sum_{i \in P_{\text{cand}}} d_{\text{Hav}}(r, \tilde{y}_i)$, and the one that minimizes this sum is selected, guaranteeing better spatial accuracy in the final trajectory. A pseudocode illustrating this optimal combination procedure is presented below.

Algorithm 1 Optimal Combination of Trajectories

Requires: Perturbed trajectories $\tilde{y}^1, \dots, \tilde{y}^n$, privacy budget ϵ , spatial domain P

Ensure: Optimized final trajectory $\hat{y}$

1: Initialize $\hat{y} \leftarrow \tilde{y}^1$

for each point $p_i \in \tilde{y}^2$ **do**

Obtain domain restricted by neighbors $P_3: i \text{ and } p_i$: $R = \{r \in P \mid d_{\text{Hav}}(r, p_i) \leq \epsilon/2\}$

r satisfies the angular constraints of p_i

4: Select $\hat{p}_i = \arg \min_{r \in R} (d_{\text{Hav}}(r, p_i) + \sum_{j \in \mathcal{N}(i)} d_{\text{Hav}}(r, \tilde{y}_j^i))$

5: Update final path $\hat{y} \leftarrow \hat{y} \cup \{\hat{p}_i\}$

6: **end for**

return $\hat{y}$

2.4.6. Analysis of the perturbation graph

Figure 2.3 illustrates the effect of the Pivot Sampling algorithm applied to a path of $N = 20$ points with a privacy budget $\epsilon = 20$. This choice of epsilon aims to maintain a direct relationship between the privacy budget and the number of perturbations performed, considering that within the framework of local differential privacy (LDP), each perturbed point consumes a portion of the total privacy budget. Thus, by assigning a ϵ proportional to the number of published coordinates, the cumulative privacy loss resulting from additive composition is mitigated, while preserving a reasonable level of spatial utility in the perturbed data. The graph shows that the perturbed points exhibit a strategic dispersion around the original path, reflecting the two-stage nature of the perturbation: independent for pivots and directional for non-pivots.

The pivot points, both even and odd depending on the path copy, experience independent perturbations through the Exponential Mechanism, resulting in more pronounced deviations from their original positions. This is observed in the irregular dispersion of some points that move significantly away from the original path. In contrast, the non-pivot points are perturbed while remaining within angular constraints determined by their neighboring pivots, generating a pattern where certain points align with specific directions, preserving the local structure of the path and avoiding abrupt distortions in the overall topology. The intersection of angular domains from multiple neighboring pivots creates a "funnel" effect that limits the perturbation space for the non-pivot points, explaining why some remain relatively close to their original position while others can move further away depending on the size of the allowed restricted domain.

The algorithm achieves a moderate balance between privacy and structure preservation with $\epsilon = 20$. Unlike the pure Laplace mechanism, where the perturbation is **isotropic**

(i.e., the noise is distributed uniformly in all directions around the original point), Pivot Sampling introduces asymmetry into the perturbation: greater freedom in unrestricted directions and greater precision in directions controlled by pivots. The final optimization process combines the two versions of the trajectory, with even and odd pivots alternately, reducing the accumulated error. This is evident in certain endpoints that are positioned between the two perturbed trajectories, minimizing the sum of Haversine distances with respect to both versions. The distinct markers (circles and crosses) for the initial and final points clearly identify the orientation and temporal order of the trajectory, confirming that the algorithm respects the sequence of points without time reversals. Furthermore, the optimal granularity [D] The granularity selected by the algorithm determines the precision with which the directions are divided: a larger granularity favors directional precision but reduces the probability of success, while a smaller granularity expands the perturbation domains but better preserves the true directions.

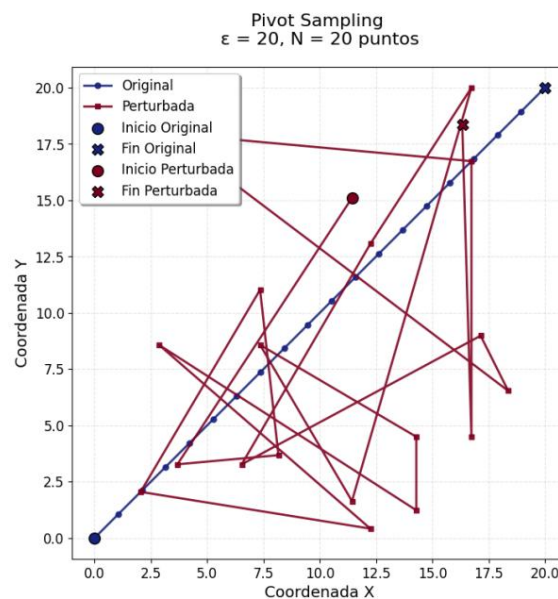


Figure 2.3: Original and perturbed trajectory using Pivot Sampling with $\epsilon = 20$ and $N = 20$ points. Circles indicate the starting points, while crosses mark the endpoints of each trajectory. The scattering reflects the pivot-independent perturbation and the directional restriction of non-pivot points.

In conclusion, the graph demonstrates how the Pivot Sampling algorithm introduces more sophisticated perturbation control compared to purely isotropic methods, leveraging directional information to preserve the spatial coherence of the trajectory while maintaining local differential privacy guarantees. This visualization allows us to understand how the strategic exchange of pivots between two copies of the trajectory and the subsequent composition optimize the balance between privacy and utility, especially in scenarios where the temporal and spatial structure of the trajectory is critical.

2.5. Anchor-Based Pivot Sampling (ATP)

The **Anchor-Based Pivot Sampling** (ATP) mechanism [12] is a **local differential privacy** (LDP)-based approach that improves traditional pivot sampling techniques by incorporating spatial anchors and adaptive perturbation domains. This allows for a reduction significantly the noise introduced into geographical trajectories when applying mechanisms of differential privacy, especially when the privacy budget ϵ is low. According to the Tobler's First Law of Geography, "everything is related to everything else, but the Things that are closer together are more related to each other." In the context of human trajectories, This implies that the points of a trajectory tend to cluster spatially into regions compact. However, traditional mechanisms, such as Pivot Sampling (TP), generate perturbations over very large domains P , which leads to a high probability of Selecting noise points far from the original path, particularly when the privacy budget ϵ is small. To mitigate this problem, ATP adaptively restricts the perturbation domain to a circular region centered on an anchor pivot, calculated for each individual trajectory. This approach allows the geographical utility of the data to be preserved while maintaining compliance with the differential privacy model.

2.5.1. Anchorage Calculation

Let $\tilde{y} = \{p_1, p_2, \dots, p_n\}$ be a path composed of geographic coordinates. It is defined an anchor $\tilde{y}$ as the average of all positions

$$\tilde{y} = \left(\frac{\sum_{i=1}^n \text{lat}(p_i)}{n}, \frac{\sum_{i=1}^n \text{lon}(p_i)}{n} \right). \quad (2.5)$$

This point $\tilde{y}$ does not necessarily belong to the discrete set of possible locations P . Therefore Therefore, the point closest to $\tilde{y}$ in P is selected as the anchor pivot $p_{\tilde{y}}$, using as metric the geodetic distance (for example, the Haversine distance)

$$p_{\tilde{y}} = \arg \min_{p \in P} \text{dist}(\tilde{y}, p). \quad (2.6)$$

The anchor pivot $p_{\tilde{y}}$ is perturbed to ensure differential privacy through the **Mechanism Exponential** (EM). This mechanism selects a location $\hat{p}_{\tilde{y}} \in P$ with probability

$$\Pr[\hat{p}_{\tilde{y}} = p] = \frac{\exp \left(\frac{\tilde{y} \times u(p_{\tilde{y}}, p)}{2\epsilon u} \right)}{\sum_{p \in P} \exp \left(\frac{\tilde{y} \times u(p_{\tilde{y}}, p)}{2\epsilon u} \right)}, \quad (2.7)$$

where the utility function favors nearby points

$$u(p_{\tilde{y}}, p) = \tilde{y} \text{dist}(p_{\tilde{y}}, p) \quad (2.8)$$

and the sensitivity $\tilde{y}_u$ is calculated as

$$\tilde{y}_u = \max_{p, \tilde{p} \in P} \text{dist}(p, \tilde{p}) - \text{dist}(p, p). \quad (2.9)$$

A high $\tilde{y}$ concentrates the probability at points close to p , while a low $\tilde{y}$ produces a more uniform distribution.

Illustrative example. Suppose the path is $\tilde{y} = \{p_1, p_2, p_3\}$ with simplified coordinates

$$p_1 = (1.00, 1.00), p_2 = (1.02, 1.03), p_3 = (0.98, 1.05).$$

The anchorage is calculated as the average

$$\bar{y} = \frac{1.00+1.02+0.98}{3}, \frac{1.00+1.03+1.05}{3} = (1.00, 1.027).$$

Given the set of locations $P = \{p_1, p_2, p_3, p_4\}$, where $\tilde{y} \in P$ and the additional point $p_4 = (1.10, 0.95)$ represents a candidate location outside the original path, the point closest to $\tilde{y}$ within P is sought.

The minimum distance is obtained for p_1 , therefore

$$p = p_1.$$

Now the Exponential Mechanism (EM) is applied with $\tilde{y} = 1$. The utility is given by $u(p, \tilde{y}) = \tilde{y} \text{dist}(p, \tilde{y})$. If we assume the following distances in kilometers

$$\text{dist}(p, p_1) = 0, \text{dist}(p, p_2) = 3, \text{dist}(p, p_3) = 5, \text{dist}(p, p_4) = 7,$$

the weights are

$$w(p_1) = e^{-0} = 1, w(p_2) = e^{-1.5} \approx 0.223, w(p_3) = e^{-2.5} \approx 0.082, w(p_4) = e^{-3.5} \approx 0.030.$$

Normalizing,

$$\Pr[\hat{p} = p_1] \approx 0.73, \Pr[\hat{p} = p_2] \approx 0.16, \Pr[\hat{p} = p_3] \approx 0.06, \Pr[\hat{p} = p_4] \approx 0.02.$$

Thus, with high probability the perturbed anchor $\hat{p}$ coincides with p_1 , although there is still a small probability of selecting other points to preserve privacy.

2.5.2. Restricted disturbance domain

Once the perturbed pivot $\hat{p}$ is obtained, it is not efficient to apply privacy mechanisms over the entire global domain P , as this can be too large and sparse. Instead, the **Anchor-Based Pivot Sampling** mechanism restricts the perturbation domain to a circular region centered at $\hat{p}$, with a radius R calibrated for each trajectory.

In this way, only points close to the disturbed pivot are considered as candidates. Formally, the restricted domain is defined as

$$\hat{P}_{\tilde{y}} = \{ p \in P \mid \text{dist}(\hat{p}_{\tilde{y}}, p) \leq R \}. \quad (2.10)$$

This set $\hat{P}_{\tilde{y}}$ contains all points P that lie inside the circumference of radius R around $\hat{p}_{\tilde{y}}$. The parameter R controls the balance between privacy and utility: A small value concentrates the points around the original path (greater utility, less privacy), while a large value includes more possible locations (greater privacy, lower utility).

Illustrative example. Continuing with the previous example, suppose the disturbed anchor is $\hat{p}_{\tilde{y}} = p_1 = (1.00, 1.00)$. Let the global set $P = \{p_1, p_2, p_3, p_4\}$ with the following distances (in km) with respect to $\hat{p}_{\tilde{y}}$

$$\text{dist}(\hat{p}_{\tilde{y}}, p_1) = 0, \text{dist}(\hat{p}_{\tilde{y}}, p_2) = 3, \text{dist}(\hat{p}_{\tilde{y}}, p_3) = 5, \text{dist}(\hat{p}_{\tilde{y}}, p_4) = 7.$$

If a radius $R = 4$ km is set, the restricted domain becomes

$$\hat{P}_{\tilde{y}} = \{ p \in P \mid \text{dist}(\hat{p}_{\tilde{y}}, p) \leq 4 \} = \{p_1, p_2\}.$$

With this value, only points p_1 and p_2 are considered candidates, reducing the dispersion of the perturbation domain. In contrast, if $R = 7$ km, then

$$\hat{P}_{\tilde{y}} = \{p_1, p_2, p_3, p_4\},$$

which increases privacy at the cost of including points further away from the original path.

2.5.3. Determination of the Adaptive Radius

Since different users generate paths with very different coverage areas, the Using a fixed radius R for all of them is inefficient. If R is too small, it can Excluding a significant portion of the original points on the trajectory reduces its usefulness. Conversely, if R is too large, numerous noisy points are incorporated into the system. candidate, degrading the accuracy of the mechanism. To mitigate this problem, a adaptive strategy, assigning a different R value to each trajectory based on its spatial characteristics.

Calculation of the maximum radius. Let $\tilde{y} = \{p_1, p_2, \dots, p_n\}$ be a path, and $\hat{p}_{\tilde{y}}$ the anchor pivot perturbed. The maximum radius associated with $\tilde{y}$ is defined as the greatest distance between $\hat{p}_{\tilde{y}}$ and a point on the trajectory

$$R_{\tilde{y}}^{\max} = \max_{p \in P_{\tilde{y}}} \text{dist}(\hat{p}_{\tilde{y}}, p), \quad (2.11)$$

where $\tilde{P} \subseteq P$ denotes the set of points that make up the trajectory. This value guarantees that all points of the trajectory are within the circular region of radius $R_{\max}$ centered on $\tilde{p}$.

Radio disturbance. To preserve privacy, $R_{\max}$ It is not used directly, but which must be perturbed. Since it is a bounded quantity, the **Square-Wave Mechanism (SW)** [12] is applied, which is designed to handle continuous data within a finite range. First, the radius is normalized with respect to the maximum possible value

$$r = \frac{R_{\max}}{\tilde{y}R}, \quad \text{where } \tilde{y}R = \max_{\tilde{p} \in \tilde{P}} \text{dist}(\tilde{p}, p).$$

Then, r is perturbed by the SW mechanism to obtain $\hat{r}$. Finally, the perturbed value- The distance is rescaled to the real domain of distances

$$R_{\max}^{\hat{r}} = \frac{(\hat{r} + b) \tilde{y}R}{2b + 1}. \quad (2.12)$$

Formally, the SW M mechanism satisfies $\tilde{y}$ -LDP if, for any input $x \in [0, 1]$ and exit $y \in [yb, b + 1]$, the probability of exit is given by

$$Pr[M(x) = y] = \begin{cases} \frac{1}{2be^{\tilde{y}} + 1}, & \text{if } |x - y| \leq b, \\ \frac{1}{2be^{\tilde{y}} + 1}, & \text{in another case.} \end{cases}$$

where

$$b = \frac{\tilde{y}e^{\tilde{y}}e^{\tilde{y}} + 1}{2e^{\tilde{y}}(e^{\tilde{y}} - 1 - \tilde{y})}.$$

The use of SW can cause the perturbed radius to be excessively small or large under small values of $\tilde{y}$ due to the high level of randomness. However, considering that the size of the human mobility regions approximately follows a distribution

In power-law type [13], the occurrence of very large radii is infrequent. Furthermore, even if

$R_{\max}^{\hat{r}}$ The resulting radius is larger than the original, which can be beneficial: the enlarged region manages to cover with greater probability the actual points of the trajectory, thus maintaining utility, while that the increase in noise is moderate compared to the total size of the space. For this reason, an **additional calibration** is used that adjusts the disturbed radii towards a “central” value when $\tilde{y}$ is small, reducing the effects of extreme variability.

Illustrative example. Let us again consider the path $\tilde{y} = \{p_1, p_2, p_3\}$ with

$$p_1 = (1.00, 1.00), p_2 = (1.02, 1.03), p_3 = (0.98, 1.05).$$

Let's assume the disturbed anchor is $\hat{p} = p_1$. 1) Calculation of the maximum radius. The distances from $\hat{p}$ to the points on the trajectory are

$$\text{dist}(p_1, p_1) = 0, \text{dist}(p_1, p_2) = 3 \text{ km}, \text{dist}(p_1, p_3) = 5 \text{ km}.$$

Therefore,

$$R_{\hat{p}}^{\max} = \max\{0, 3, 5\} = 5 \text{ km}.$$

2) Normalization. If the maximum distance in the domain is

$$\max_{p \in P} \text{dist}(\hat{p}, p) = 7 \text{ km}, \tilde{R} = m$$

so

$$r = \frac{R_{\hat{p}}^{\max}}{\tilde{R}} = 5 \div 7.14 = 0.7$$

3) Disturbance with SW. Let $\tilde{y} = 1$. The parameter b is calculated as

$$b = \frac{\tilde{y}e^{\tilde{y}} - eb^{\tilde{y}} + 1}{2e^{\tilde{y}}(e^{\tilde{y}} - 1 - \tilde{y})}} = \frac{1 \times e - e + 1}{2e(e - 2)}.$$

With $e \approx 2.718$, we obtain

$$b \approx \frac{0.718}{2 \times 2.718 \times 0.718} \approx 0.184.$$

Assuming that the SW mechanism returns a perturbed value $\hat{r} = 0.60$, the radius is rescaled

to

$$R_{\hat{p}}^{\max} = \frac{(\hat{r} + b) \tilde{R} - 2b}{1} = \frac{(0.60 + 0.184) \times 7}{1} \approx 4.01 \text{ km}.$$

Interpretation. Although the actual maximum radius was 5 km, the SW mechanism produces a perturbed value of ≈ 4 km. In this way, most of the utility is preserved (the region still covers almost all points of the trajectory), while introducing enough randomness to guarantee differential privacy.

2.5.4. Radius Calibration

The calibration of the radius consists of two main steps (i) determining the calibration center- (ii) calculate the size of the adjustment step as a function of $\tilde{y}$ and calibrate the radius. Recall that $R_{\max}$ is bounded and corresponds to the furthest distance between the disturbed anchor point $\hat{p}$ and the points on the trajectory. To identify a suitable value $\tilde{y}$ as the calibration center, we seek to amplify the impact of the points that satisfy a possible range of maximum distances from $\hat{p}$. Since the input domain of the SW mechanism is $[0, 1]$, trial values are generated uniformly to avoid privacy leaks and calculate the possible ranges. Let

$$V = \{vk \mid vk = 0.1 \times k, k = 0, 1, \dots, 10\}$$

the set of test values, and are defined

$$l_k = v_k - b, u_k = v_k + b.$$

Thus, the set of relevant values VR is determined as

$$VR = \{v_k \in V \mid l_k \leq b \leq u_k, 0 \leq k \leq \frac{(2b+1)R^{\max}}{\bar{y}_R} - 1\}.$$

From this, the lower and upper ranges are obtained.

$$l = \min v_k, u = \max v_k, \forall k \in VR.$$

The set of candidate points is defined as

$$P^R = \{p \in P \mid l \leq b \leq u, \bar{y}_R \frac{(2b+1) \text{dist}(\hat{p}, p)}{e^{\bar{y}}}\}$$

The values in V are used only as tests. For each v_k , the probability is calculated.

$$q = \frac{e^{\bar{y}}}{2be^{\bar{y}} + 1},$$

which represents the fraction of probabilistic mass concentrated in the most probable output range of the SW mechanism. Finally, the **calibration center** $\bar{y}$ is calculated as the weighted average of the distances, assigning greater weight to candidate points

$$\bar{y} = \frac{p \bar{y}_R q \times \text{dist}(\hat{p}, p) + \sum_{p \in P^R} p \bar{y}_R (1 - q) \times \text{dist}(\hat{p}, p)}{PR + (1 - q)|P^R|}.$$

Illustrative example. Continuing with the previous case, let's assume that the calculated perturbed radius was

$$R^{\max} = 4.01 \text{ km}, \bar{y}_R = 7 \text{ km}, \bar{y} = 1,$$

And recall that the parameter of the SW mechanism was $b \approx 0.184$. 1) Determination of the relevant values. The normalization factor is

$$z = \frac{(2b+1)R^{\max}}{\bar{y}_R} b = \frac{1.368 \times 4.01}{7} \approx 0.184 \approx 0.598.$$

Let $V = \{0, 0.1, 0.2, \dots, 1.0\}$. For $v_6 = 0.6$, it holds that

$$l_6 = 0.6 - 0.184 = 0.416, u_6 = 0.6 + 0.184 = 0.784,$$

and indeed $z = 0.598 \in [0.416, 0.784]$. Therefore, $VR = \{0.6\}$ and we obtain $l = u = 0.6$.

2) Candidate set. The set P^R includes those points $p \in P$ that satisfy

$$\bar{y} \leq b \leq u, \bar{y}_R \frac{(2b+1) \text{dist}(\hat{p}, p)}{e^{\bar{y}}}$$

In this case, when evaluating the distances with respect to $\hat{p} = p_1$, it is found that p_2 (3 km) and p_3 (5 km) fall within the allowed range, therefore

$$P^R = \{p_2, p_3\}.$$

3) Calculation of the calibration center. The weight of the SW mechanism is

$$q = \frac{\frac{\hat{y}}{2be\hat{y} + 1}}{\times 0.184 \times 2.718 + 1} = \frac{2.718}{\times 0.184 \times 2.718 + 1} \hat{y} 0.881. 2 \quad (2.13)$$

Applying the formula for the calibration center

$$P \hat{y} = \frac{q \times (\text{dist}(p_1, p_2) + \text{dist}(p_1, p_3)) + (1 - q) \times \text{dist}(p_1, p_4)}{q|P^R| + (1 - q)}. \quad (2.14)$$

Substituting the numerical values

$$\hat{y} = \frac{0.881 \times (3 + 5) + 0.119 \times 7 \times 0.881}{2 + 0.119 \times 1} = \frac{7.048 + 0.833}{1.762 + 0.119} = \frac{7.881}{1.881} \hat{y} 4.19 \text{ km}.$$

Interpretation. The calibration center $\hat{y} \hat{y} 4.19$ km adjusts the perturbed value of the radius towards a central zone, smoothing the variability introduced by the SW mechanism and maintaining a compromise between privacy and utility.

2.5.5. Adjustment of the Disturbed Radius

Once the calibration center $\hat{y}$ has been obtained, the next step is to determine how to "attract" the perturbed radius $\hat{R}^{\max}$ towards this center. Due to the property of the SW mechanism, the perturbed value becomes more concentrated as the privacy budget $\hat{y}$ increases. Therefore, the calibration effect must decrease as $\hat{y}$ grows. The calibrated radius is defines as

$$\hat{R} = \hat{R}^{\max} + \hat{y}e^{\hat{y}}, \quad (2.15)$$

where the calibration term is composed of two factors: (i) the decay factor, which controls the effect of the calibration by reducing it when $\hat{y}$ is larger, and (ii) the displacement factor $\hat{y}$, which depends on the difference between the perturbed radius and the calibration center. The displacement factor is defined as

$$1 - \hat{y} = (\hat{y} - \hat{R}^{\max}) \times \frac{1}{e^{\hat{y}/2}}, \quad (2.16)$$

where the sign of $\hat{y}$ indicates the direction of adjustment: it will be positive if the perturbed radius is less than $\hat{y}$, and negative if it is greater. In this way, the value of $\hat{y}$ does not become a constant or an extreme value. Finally, to normalize the difference between the perturbed radius and

$\tilde{y}$ in the range $[0, 1]$, the parameter $\tilde{y}$ is introduced, defined as

$$\tilde{y} = \begin{cases} \frac{\tilde{y} - \tilde{y} R^{\max}}{\tilde{y}}, & \text{if } R^{\max} \leq \tilde{y}, \\ -\frac{R^{\max} - \tilde{y}}{\tilde{y} R^{\max}}, & \text{otherwise.} \end{cases} \quad (2.17)$$

Subsequently, $\tilde{y}/2$ is used as the input to a sigmoid activation function, which is defined as

$$\tilde{y}(x) = \frac{1}{1 + e^{\tilde{y}x}},$$

introducing a nonlinear property and preventing the movement step from taking exactly the value $|\tilde{y} - R^{\max}|$. In this case, the input is $x = \tilde{y}/2$, so the applied sigmoid is

$$\tilde{y} \frac{\tilde{y}}{2} = \frac{1}{1 + e^{\tilde{y}^2/2}}.$$

Thanks to this calibration process, the perturbed radius is adjusted in a controlled manner, and the set of perturbed points of the input path is finally obtained.

Illustrative example. Suppose that, after the disturbance and intermediate calibration, it obtained

$$R^{\max} = 4.01 \text{ km}, \tilde{y} = 4.19 \text{ km}, \tilde{y}R = 7 \text{ km}.$$

1) Calculation of $\tilde{y}$. Since $R^{\max} \leq \tilde{y}$, the first definition applies

$$\tilde{y} = \frac{\tilde{y} - \tilde{y} R^{\max}}{\tilde{y}} = \frac{4.19 - 4.01}{4.19} \approx 0.043.$$

2) Calculation of the displacement factor $\tilde{y}$.

$$\tilde{y} = (\tilde{y} - R^{\max}) \times \frac{1}{e^{\tilde{y}^2/2}}.$$

Evaluating

$$\tilde{y} = 0.18, \tilde{y}/2 \approx 0.0215, e^{\tilde{y}^2/2} \approx 0.979, \tilde{y} - R^{\max}$$

Therefore

$$\tilde{y} = 0.18 \times \frac{1}{e^{0.979}} = 0.18 \times \frac{1}{0.021} \approx 8.57.$$

3) Calculation of the calibrated radius. For different values of $\tilde{y}$, the effect of the decay $e^{-\tilde{y}^2/2}$ will be

$$R = R^{\max} + \tilde{y}e^{\tilde{y}^2/2}.$$

- If $\tilde{y} = 0.5$

$$R = 4.01 + 8.57 \times e^{-0.5^2/2} \approx 4.01 + 5.20 = 9.21 \text{ km}.$$

- If $\tilde{y} = 1$

$$\hat{R} = 4.01 + 8.57 \times e^{-1} \tilde{y} = 4.01 + 3.15 = 7.16 \text{ km.}$$

- If $\tilde{y} = 2$

$$\hat{R} = 4.01 + 8.57 \times e^{-2} \tilde{y} = 4.01 + 1.16 = 5.17 \text{ km.}$$

Interpretation. When $\tilde{y}$ is small, the calibration effect is strong and the radius expands significantly (higher privacy, lower utility). As $\tilde{y}$ grows, the term e decays and the calibrated radius approaches $\hat{R}_{\max} \tilde{y}$, better preserving utility.

2.5.6. Analysis of the perturbation graph

Figure 2.4 shows the effect of the Anchor-based Pivot Sampling (ATP) mechanism applied to a diagonal path of $N = 20$ points with a privacy budget $\tilde{y} = 20$.

This mechanism combines directional anchoring principles and constrained pivots to control noise dispersion based on the local path structure. Unlike purely isotropic mechanisms, ATP introduces a directional bias that favors the preservation of the overall shape and the spatial continuity of segments between adjacent points.

The graph shows that the disturbed points (red line) maintain a general tendency close to the original trajectory (blue line), although with notable deviations in intermediate regions where the trajectory has a lower density of effective anchors. The use of $\tilde{y} = 20$ implies a relatively low level of privacy, which translates into a bounded dispersion of the noise. However, the mechanism maintains the local independence of the disturbances and, at the same time, avoids extreme distortions thanks to the geometric constraints imposed by the pivots. This is reflected in the fact that, despite the lateral deviations, the temporal sequence of the points is preserved without inversions, and the general direction of advance of the diagonal trajectory remains recognizable. The initial and final markers (circles and crosses) allow us to identify the orientation of both trajectories, showing that ATP maintains topological coherence and respects the temporal structure of the movement.

Unlike the Laplace mechanism, where scattering is **isotropic**—that is, the noise is distributed uniformly in all directions around the original point—ATP applies an **anisotropic perturbation**, in which the magnitude and direction of the noise depend on the geometric context and the reference pivots. This characteristic allows the perturbation to be directed in less sensitive directions and limited in those that affect spatial coherence.

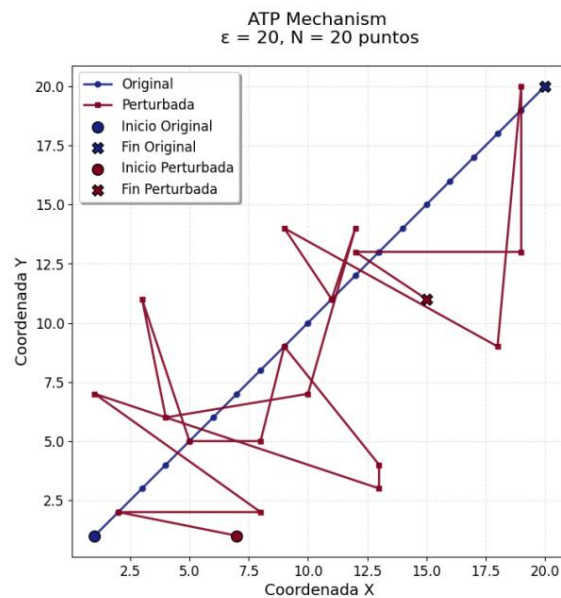


Figure 2.4: Original and perturbed trajectory by the ATP mechanism with $\epsilon = 20$ and $N = 20$ points. The circles indicate the initial points and the crosses the final points of each trajectory.

In conclusion, the graph shows that the ATP mechanism achieves an effective balance between privacy and utility, preserving the overall shape of the trajectory while introducing localized noise that protects individual positions. Anchor-guided anisotropic perturbation maintains spatial continuity and directional coherence, offering a significant improvement over purely random perturbation methods.

Chapter 3

Analysis of the Behavior of Privacy Budget

As explained in Section 2.1, central differential privacy (CDP) mechanisms and local differential privacy (LDP) differ in the way they manage the budget of privacy ϵ . In this section, the role of this parameter within the context is explored in greater depth. experimental, highlighting how its configuration affects the utility and noise level in the disturbed trajectories.

The privacy budget ϵ is a fundamental parameter in differential privacy mechanisms that determines the level of protection offered to individuals. As mentioned previously, small values of ϵ guarantee greater privacy through addition significant noise, while large values allow for greater precision in perturbed data at the cost of reduced privacy. This parameter is not static and its behavior It varies depending on the privacy approach used.

In particular, algorithms based on **local differential privacy** (LDP), such as Pivot Sampling and LDPTTrace require that the privacy budget be distributed among all the points of the trajectory [9]. This means that ϵ must be adjusted according to the quantity of points N on the trajectory, since each point consumes part of the overall budget. Formally, in LDP, when multiple points participate in the perturbation process, the The total budget is divided or applied through composition mechanisms, affecting the relationship between ϵ and the level of noise introduced.

Conversely, the **Laplace Mechanism** implements **central differential privacy** (CDP), where there is a trusted curator who has access to all the data. In this , In this scenario, the privacy budget ϵ is global and does not depend on the number of points in N the path. The parameter ϵ is applied uniformly to each point, without needing to composition, allowing a more direct and homogeneous application of Laplace noise.

3.1. Privacy Budget Analysis Methodology

To understand the behavior of these algorithms and quantify the trade-off between Privacy and utility: a comparative analysis is performed using an actual route from Beijing (Beijing) containing $N = 908$ points. This trajectory of significant dimensionality allows us to observe how the different mechanisms behave in the face of real high-resolution data. dimensionality.

3.1.1. Evaluation Metrics

The analysis is performed using the following metrics:

Table 3.1: Metrics used to evaluate perturbed trajectories.

Metrics	Description
Mean Distance	Average of the Haversine distances between original points and their corresponding versions disturbed. Quantifies the average displacement introduced by the mechanism.
Maximum Distance	Maximum observed displacement between any pair of original and disturbed points. Indicates the worst case of disturbance.
Standard Deviation	Measure of variability in displacements. A high deviation indicates disturbances inconsistent.
Length Error	Difference between the total length of the original and disturbed paths, calculated as the sum of Haversine distances between consecutive points.
Directional Error	Measure of change in the general direction of movement between the original and perturbed trajectory, based on the angle between successive vectors.
Processing Time: Total time required to execute the perturbation algorithm on the trajectory.	Indicator of computational efficiency.

3.1.2. Data Set

For the analysis, a real trajectory from the city of Pek'yn (Beijing) was used, originating from the **Microsoft Geolife GPS trajectory dataset** [14, 15, 16].

This collection, compiled by Microsoft Research, contains more than 17,000 trajectories recorded by 178 users between 2007 and 2011, representing real mobility patterns urban. The selected trajectory includes $N = 908$ points with spatial and temporal variability, suitable for evaluating the performance of disturbance mechanisms in real data high resolution.

3.1.3. Normalization of Epsilon to a Unified Scale

Since algorithms employ different interpretations of the privacy budget ϵ and may require significantly different numerical values to achieve comparable levels of privacy, it is necessary to normalize the values of ϵ to a common scale to facilitate comparison. A unified scale of 1 to 5 is established, where:

Table 3.2: Unified scale of normalization of the privacy budget ϵ .

Interpretation Level	
1	Maximum privacy (minimum utility)
2	High privacy (low utility)
3	A moderate balance between privacy and utility
4	Low privacy (high utility)
5	Minimum privacy (maximum utility)

This normalization allows for fair comparisons between the three algorithms, compensating for the inherent differences in how each interprets and applies the budget. The specific mapping of original ϵ values to this unified scale is described individually for each algorithm in the subsequent sections. Next, we will present a detailed analysis of the behavior of each perturbation algorithm under different levels of the unified privacy scale, using the actual Peking trajectory with $N = 908$ points as a basis for evaluation. For each algorithm, the mapping of its original parameters to the scale of 1 to 5 will be specified, as well as the results obtained in terms of the metrics. evaluation described above.

3.2. Laplace Mechanism

As mentioned in the chapter introduction, the unified privacy scale (1 to 5) was established to facilitate fair comparisons between algorithms. However, the Laplace Mechanism presents a particular challenge: it cannot directly use the values of ϵ specified on that scale because their parameters depend critically on the numerical scale of the input coordinates.

Since this analysis works with real geographic coordinates (latitude and longitude) converted to a local metric system, it is essential to make a specific adjustment of the privacy budget. This adjustment is made through the concept of **average radius of desired disturbance** in meters, which is an interpretable physical quantity. From this The desired radius is calculated, and the specific parameter ϵ that must be used for the Mechanism is determined. Laplace's.

The following subsections detail how this adjustment is made and why it is necessary for to ensure that the behavior of the Laplace Mechanism is comparable to that of the others algorithms on the unified scale.

3.2.1. Importance of the Coordinate Scale

The Laplace Mechanism is agnostic with respect to the **physical units** of input (meters, kilometers, degrees, etc.), but its behavior depends critically on the **numerical scale** in which the coordinates are expressed. This distinction is fundamental and explains why it is necessary to adjust the parameter $\tilde{y}$ even after converting geographic coordinates to a local metric system. The relation that governs the Laplace mechanism in 2D is $E[r] = \frac{2}{\tilde{y}}$.

where $E[r]$ is the expected mean displacement distance and $\tilde{y}$ is the parameter of Privacy. This formula is exact and does not depend on units, but its practical interpretation $\tilde{y}$ depends on the scale of the input coordinates.

Small-scale problem (degrees latitude/longitude). If Laplace's equation is applied directly to Geographic coordinates in degrees:

- Typical coordinate range: 0 to 180 degrees
- If $\tilde{y} = 0.01$ (small arbitrary value) is used : ~

$$E[r] = \frac{2}{\tilde{y}} = 200 \text{ degrees} \quad (3.1)$$

- Interpretation: The average noise is 200 degrees, equivalent to approximately 22,200 km
- Result: the disturbance is completely disproportionate and uncontrollable

The problem is not with the mechanism, but with the scale: a small $\tilde{y}$ produces large perturbations in the chosen unit, which is sensible when the scale is large (meters) but absurd when it is small (degrees).

Solution: Projection to local coordinates in meters. Using equirectangular projection of the first point of the trajectory as a reference, the lat/lon coordinates are converted to a local system (x, y) expressed in meters:

$$x = (\tilde{y} - \tilde{y}_0) \times \cos(\tilde{y}_0) \times R, \quad y = (\tilde{y} - \tilde{y}_0) \times R, \quad (3.2)$$

where $R \approx 6,371,000$ m is the Earth's radius. After this conversion:

- Typical coordinate range: 0 to 100,000 meters (depends on the length of the trajectory)
- With the same $\tilde{y} = 0.01$:

$$E[r] = \frac{2}{0.01} = 200 \text{ meters} \quad (3.3)$$

- Interpretation: the average noise level is 200 meters (realistic and controllable)
- Result: the disturbance is proportional to the scale of the data

Key conclusion: The same value of $\tilde{y}$ produces completely different results because the scale of the coordinates changed. Therefore, $\tilde{y}$ must be selected according to the scale specific numerical data.

3.2.2. Selection of Privacy Parameters using Desired Medium Radius

Instead of choosing $\tilde{y}$ arbitrarily, it is more interpretable to directly specify the **Determine the desired mean radius of disturbance** in meters and calculate the corresponding $\tilde{y}$. The relationship The reverse is:

$$\tilde{y} = \frac{2}{\text{Desired Medium Radius (m)}}.$$

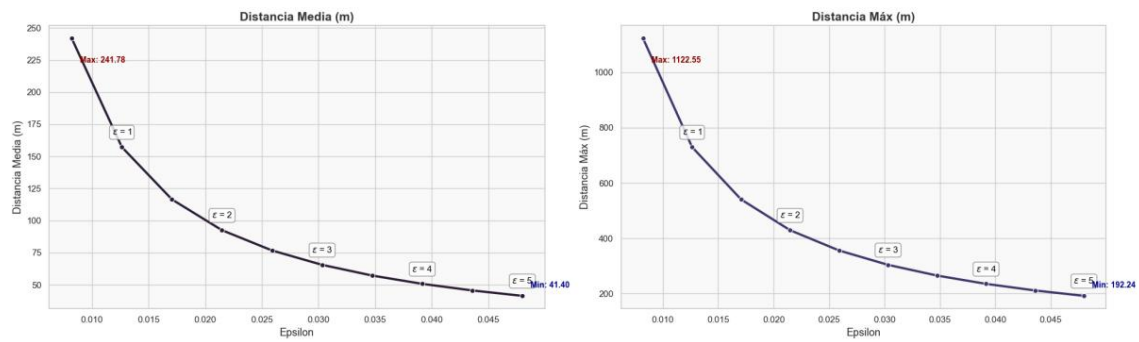
For this analysis, five levels of privacy were established based on comparisons with The results of Pivot Sampling and LDPTrace. The desired mean radii in meters were: The values of $\tilde{y}$ were calculated automatically using the formula $\tilde{y} = \frac{2}{r_{\text{medio}}}$:

Table 3.3: Mapping of privacy levels of the Laplace Mechanism: desired average radius, calculated $\tilde{y}$ values , and unified scale.

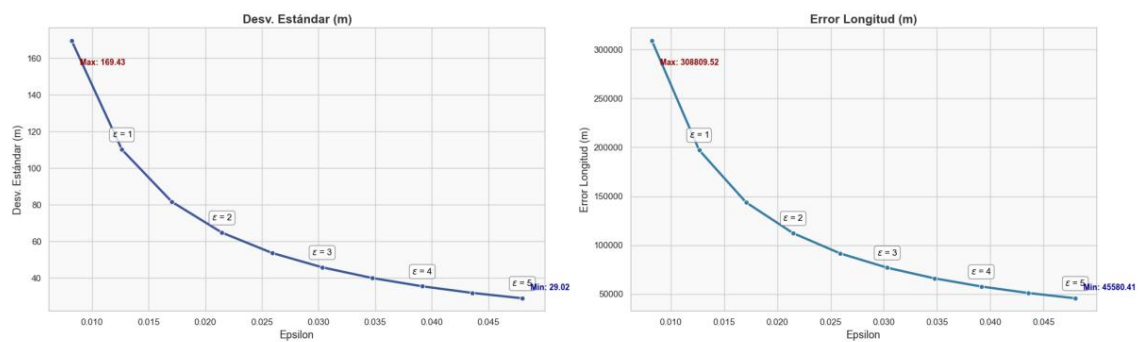
Desired	Average Radius Level (m) $\tilde{y} = 2/r_{\text{medio}}$	158.0 93.5	Interpretation
1		0.01264	Maximum privacy
2		0.02148	High privacy
3	65.9	0.03032	Moderate balance
4	51.1	0.03916	Low privacy
5	45.9	0.04358	Minimal privacy

3.2.3. Evaluation of Metrics

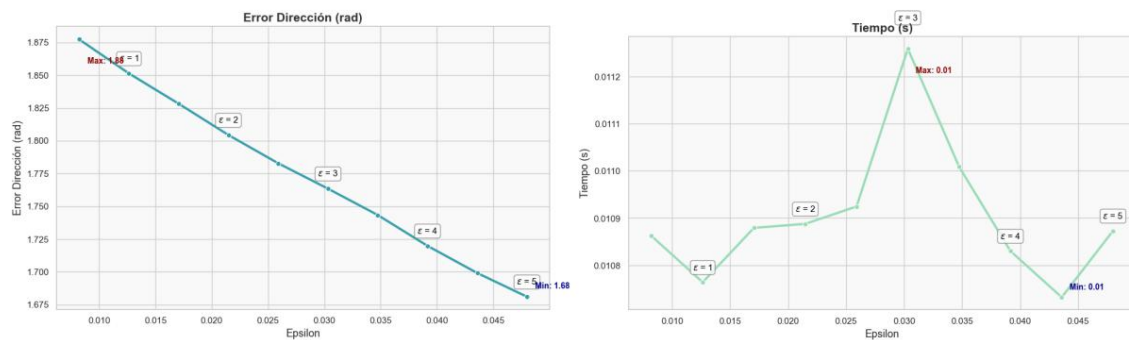
The results of the Laplace Mechanism applied to the actual Peking trajectory with $N = 908$ points are presented below, evaluated using the seven metrics described in the Chapter introduction.



(a) Average Distance (m) vs. Privacy Budget ϵ . (b) Maximum Distance (m) vs. Privacy Budget ϵ .



(c) Standard Deviation (m) vs. Privacy Budget ϵ . (d) Length Error (m) vs. Privacy Budget ϵ .



(e) Direction Error (rad) vs. Privacy Budget ϵ . (f) Processing Time (s) vs. Privacy Budget ϵ .

Figure 3.1: Error and performance metrics of the Laplace Mechanism under different values of ϵ .

Average Distance. The average distance between original and perturbed points exhibits an inversely proportional relationship with ϵ . As expected theoretically, with $\epsilon = 0.01264$ (level 1) an average distance of 241.78 meters is obtained, while with $\epsilon = 0.04358$ (Level 5) decreases to 41.40 meters. This decreasing trend is consistent with the formula $E[r] = \frac{1}{\epsilon}$, although the observed values are higher than theoretically expected (79.1 m and 23.0 m respectively), which indicates the presence of variability inherent in nature stochastic mechanism.

Maximum Distance. The maximum distance exhibits significantly greater variability than the average distance, ranging from 1122.55 meters at level 1 to 192.24 meters at Level 5. This metric illustrates that although the average noise is controlled, there are points Individuals experiencing extreme perturbations, which is characteristic of the exponential distribution of the Laplace mechanism. The ratio between maximum and average distance It is approximately 4.7 at level 1 and 4.6 at level 5, showing consistency in this tailing behavior.

Standard Deviation. The standard deviation of the radial displacement follows a trend similar to the average distance, decreasing from 169.43 meters (level 1) to 29.02 meters (level 5). The relationship between standard deviation and mean distance remains approximately constant (≈ 0.70), confirming the homogeneity of the disturbance pattern across different levels of privacy.

Length Error. The length error is the metric that experiences the greatest variability. starting from 308,809.52 *meters* at level 1 and decreasing to 45,580.41 *meters* at the Level 5. This magnitude results from the accumulation of small errors at each point along the 908 points of the trajectory. When each displacement has random noise And independently, length errors tend to accumulate quadratically, explaining the high sensitivity of this metric to changes in ϵ .

Direction Error. The direction error measured in radians is relatively small and consistent, ranging from 1.86 radians (level 1) to 1.68 radians (level 5). The reduction is more moderate compared to other metrics, suggesting that the Laplace mechanism, when perturbing coordinates isotropically better preserves the overall orientation of the trajectory than metric aspects such as length or specific distances. This behavior is advantageous for applications where path topology is more important than metric accuracy Exactly.

Processing Time. The processing time is extremely efficient, ranging from 0.0108 to 0.0112 seconds for the entire 908-point path.

It shows small fluctuations around this average value without a clear correlation with ϵ , indicating that the computational cost is practically independent of the budget of Privacy. For a trajectory of this size, the mechanism is essentially instantaneous. from a practical perspective.

Comparative Analysis of Behavior

The Laplace Mechanism exhibits behavior highly consistent with its theoretical foundation. First, the expected relationship between the privacy parameter and the magnitude of the mean displacement, expressed as $E[r] = \frac{1}{\epsilon}$. Although variations are observed due to the stochastic nature of the sampling process,

The general trend demonstrates a consistent theoretical predictability that allows anticipating the impact of different privacy settings.

Regarding the trade-off between privacy and utility, the results show a monotonic correlation between the increase in $\tilde{y}$ and the decrease in error metrics.

This behavior demonstrates a clear and stable trade-off, which facilitates the calibration of the mechanism, according to the privacy needs of the Analysis. Thus, an increase in privacy budget, produces a proportional reduction of noise, allowing one to obtain trajectories with greater spatial fidelity.

Furthermore, the disturbance generated by the mechanism is isotropic, that is, it is distributed uniformly in all directions. This characteristic favors the preservation of the directional structure of the trajectories, maintaining the overall topological coherence even if the metric precision is altered. Thanks to this property, the Laplace Mechanism achieves maintain trajectories with shapes and orientations similar to the originals, even under low levels disturbance highs.

Another relevant aspect is the accumulation of errors in aggregate metrics, such as length total path length. Since each point is perturbed independently, the errors They accumulate proportionally to the number of points, generating more noticeable deviations in long trajectories. This characteristic is inherent to independent noise mechanisms and should be considered when analyzing data with high spatial density.

Finally, the computational performance of the Laplace Mechanism is outstanding. The processing time remains practically constant, on the order of milliseconds, regardless of the value of $\tilde{y}$ or the size of the path. This efficiency confirms that the mechanism is suitable for large-scale applications or scenarios requiring real-time disturbances, without introducing significant overhead in processing.

3.3. Pivot Sampling

Unlike the Laplace Mechanism, which implements central differential privacy (CDP), Pivot Sampling is an algorithm based on **local differential privacy** (LDP). This fundamental distinction has direct implications for how the privacy budget $\tilde{y}$.

3.3.1. Composition in Local Differential Privacy

In the context of local differential privacy (LDP), each point of a path is disturbs independently, which implies that the total privacy budget must consider the additive composition of all individual disturbances. Instead of applying To directly apply the rule $\tilde{y}_{\text{total}} = N \times \tilde{y}_{\text{point}}$, it was decided to introduce an adjustment parameter k , which It takes values between 1 and 5, to control the effective magnitude of the budget used. In this way, the privacy budget is defined as:

$$\tilde{y}_{\text{effective}} = k \times N, \quad (3.4)$$

where N represents the path length. The inclusion of the parameter k responds to the need to capture more realistically the cumulative effect of composition without excessively restricting the range of $\tilde{y}$. In practice, a fixed value of $\tilde{y}$ per point would lead to a significant loss of utility when N is large, because the additive sum including this in the budget would create an excessively strict privacy guarantee. By using k as an adjustment factor, the budget can be scaled in a controlled manner, allowing to observe how the balance between privacy and fidelity of the trajectory varies. This approach provides a flexible framework for comparative analysis: the different values of k represent scenarios with different degrees of added protection. Thus, $k = 1$ can be interpreted as a conservative composition (high privacy, lower utility), while $k = 5$ reflects a more permissive scenario that prioritizes spatial accuracy. In short, the use of the parameter k allows managing privacy composition across extensive trajectories without distorting extreme usefulness of disturbed data.

3.3.2. Unified Scale Parameter Mapping

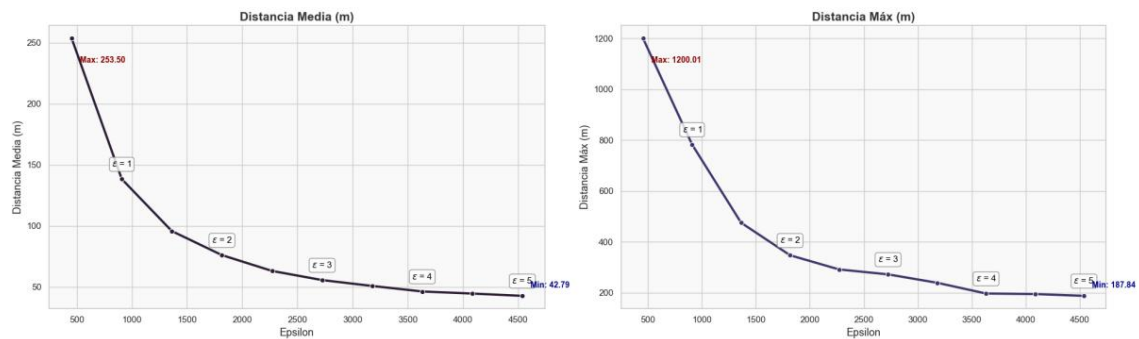
For the five levels of the unified scale (1 to 5), the following values of k are used :

Table 3.4: Mapping of privacy levels for Pivot Sampling: values of k , effective epsilon calculated, and unified scale.

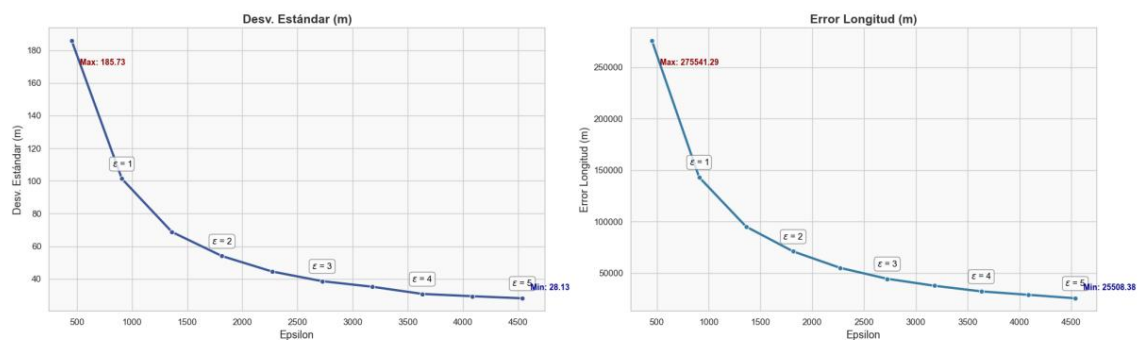
Level	k	$\tilde{y}^{\text{effective}} = k \times N$	Interpretation
1	1	$1 \times 908 = 908$	Maximum privacy
2	2	$2 \times 908 = 1816$	High privacy
3	3	$3 \times 908 = 2724$	Moderate balance
4	4	$4 \times 908 = 3632$	Low privacy
5	5	$5 \times 908 = 4540$	Minimum privacy

3.3.3. Evaluation of Metrics

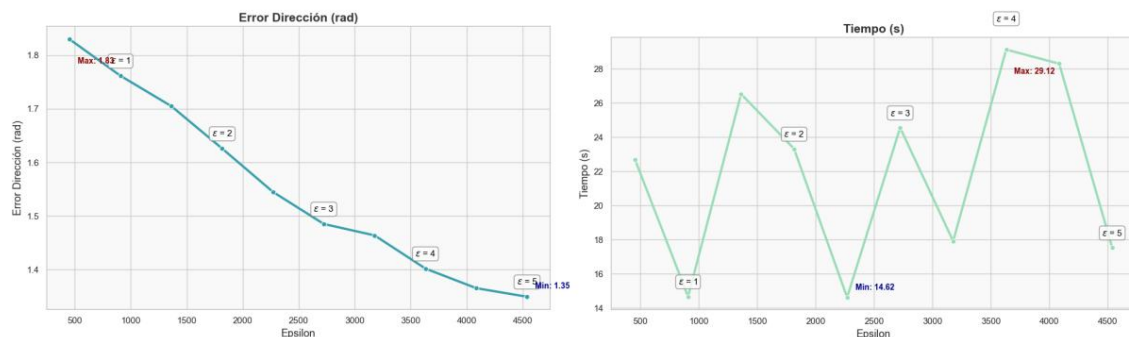
The results of the Pivot Sampling algorithm applied to the following are presented Beijing's actual trajectory with $N = 908$ points, evaluated using the seven metrics described in the chapter introduction. The values of $\tilde{y}^{\text{effective}}$ correspond to: level 1 ($\tilde{y}^{\text{eff}} = 908$), level 2 ($\tilde{y}^{\text{eff}} = 1816$), level 3 ($\tilde{y}^{\text{eff}} = 2724$), level 4 ($\tilde{y}^{\text{eff}} = 3632$), and level 5 ($\tilde{y}^{\text{eff}} = 4540$).



(a) Average Distance (m) vs. Effective Privacy Budget Efficiency . (b) Maximum Distance (m) vs. Effective Privacy Budget Efficiency .



(c) Standard Deviation (m) vs. Effective Privacy Budget Efficiency . (d) Length Error (m) vs. Effective Privacy Budget Efficiency .



(e) Direction Error (rad) vs. Effective Privacy Budget Efficiency . (f) Processing Time (s) vs. Effective Privacy Budget Efficiency .

Figure 3.2: Error and performance metrics of the Pivot Sampling algorithm under different values of the effective privacy budget ϵ .

Average Distance. The average distance between original and perturbed points exhibits an inversely proportional relationship with ϵ . With $\epsilon = 908$ (level 1) a distance is obtained average of 253.50 meters, while with $\epsilon = 4540$ (level 5) it decreases to 42.79 meters. This trend is consistent with expected behavior: larger privacy budgets (smaller ϵ , or equivalently smaller k) produce larger disturbances.

The decrease is approximately logarithmic, typical of privacy mechanisms where the benefit of additional privacy decreases with larger budgets.

Maximum Distance. The maximum distance varies between 1200.01 meters at level 1 and 187.84 meters at level 5. Compared to Laplace, Pivot Sampling produces similar maxima, although with slight variations. The ratio between maximum and average distance is approximately 4.73 at level 1 and 4.39 at level 5. This metric reflects that, although the average disturbance is controlled, there are individual points that experience significant deviations, a characteristic common to both algorithms but with different intensities depending on the privacy level.

Standard Deviation. The standard deviation of the displacement decreases from 185.73 meters (level 1) to 28.13 meters (level 5). The ratio to the mean distance is approximately 0.73 at both ends, similar to that observed in Laplace (0.70). This consistency suggests that both algorithms maintain homogeneous variability patterns across different privacy levels, although with slightly higher magnitudes in Pivot Sampling.

Length Error. The length error is again the most sensitive metric, ranging from 275,541.29 meters at level 1 to 25,508.38 meters at level 5. Although these values are similar to those of Laplace, the reduction is more pronounced in Pivot Sampling, suggesting that directional perturbation at non-pivot points partially mitigates the accumulation of length errors. The two-stage structure of the algorithm (independent pivots + directional non-pivots) appears to produce a slightly better balance than Laplace's pure isotropic perturbation in terms of length preservation.

Direction Error. The direction error varies between 1.83 radians (level 1) and 1.35 radians (level 5), showing a more significant reduction than in Laplace (where it was from 1.86 to 1.68 radians). This difference is notable: Pivot Sampling better preserves the overall orientation of the trajectory, which is a direct result of the directional perturbation of non-pivot points. By maintaining angular constraints through information from neighboring pivots, the algorithm better preserves the overall movement directions, which is a clear advantage over Laplace in applications where directional topology is important.

Processing Time. The processing time shows significant variability, ranging from approximately 14.62 to 29.12 seconds, with a non-monotonic pattern. This contrasts dramatically with Laplace, which requires less than 0.012 seconds. The reason lies in the computational complexity of the algorithm: Pivot Sampling performs multiple iterations (creation of two copies, independent perturbation of pivots, directional perturbation of non-pivots, selection of perturbation domain through intersections, and final composition). Additionally, the time increases at intermediate levels (especially level 4) due to the increasing complexity of angular domain calculations and optimizations. This is a significant trade-off: Pivot Sampling offers better preservation of structure (especially directional), but at a significant computational cost.

3.3.4. Comparative Analysis of Behavior

The Pivot Sampling algorithm exhibits a distinct behavior compared to the Laplace mechanism due to its two-stage perturbation structure. While Laplace applies a uniform isotropic perturbation in all directions, Pivot Sampling differentiates between pivot and non-pivot points: the former are perturbed independently, while the latter are perturbed under directional constraints based on their neighboring pivots. This strategy introduces a more structured perturbation that better preserves the spatial coherence of the trajectory, maintaining the overall orientation of the displacement even at high privacy levels.

In terms of the relationship between privacy and utility, Pivot Sampling reveals a particular equilibrium derived from composition in the local differential privacy (LDP) model. Although the effective budget $\epsilon_{\text{eff}} = k \times N$ is numerically larger, the practical effect on the error metrics is comparable to that of Laplace. This demonstrates that local mechanisms, operating independently at each point, require different calibrations to achieve distortion levels equivalent to those of central mechanisms.

A clear advantage of the Pivot Sampling approach lies in its ability to preserve the directional orientation of the trajectories. The average angular error remains lower than in the case of the Laplace mechanism (1.35 vs. 1.68 radians at level 5), indicating better preservation of the motion topology. This property is particularly valuable in applications where the shape and direction of the path are more relevant than absolute metric accuracy, such as in mobility analysis or transportation patterns.

Furthermore, the algorithm demonstrates a slight mitigation in the accumulation of errors of length. The partially correlated perturbation between non-pivot points reduces the quadratic growth of typical noise in independent mechanisms, generating trajectories with total distances closer to the original. This behavior suggests that the directional structure not only preserves orientation, but also some global metric consistency.

However, this improvement in spatial coherence comes at a considerable computational cost. Pivot Sampling requires between 15 and 29 seconds per trajectory, in contrast to the practically instantaneous times of the Laplace mechanism (on the order of 0.01 seconds). This increase is due to the complexity of the process, which includes pivot selection, conditional perturbations, definition of angular domains, and final composition. Furthermore, the execution time does not follow a monotonic pattern: some intermediate levels of k increase the computational load due to more demanding geometric adjustments in the definition of directional restrictions.

Overall, Pivot Sampling stands out as a local differential privacy mechanism capable of offering better structural preservation of trajectories, albeit at the expense of greater operational complexity. Its design makes it particularly suitable for decentralized environments, where users perturb their own data before sharing it, maintaining local and autonomous control over their privacy.

3.4. Anchor-Based Pivot Sampling (ATP)

Unlike traditional Pivot Sampling and the Laplace Mechanism, **Anchor-Based Pivot Sampling** (ATP) introduces an adaptive approach that restricts the disturbance domain using spatial anchors. This algorithm implements **local differential privacy**.

(LDP) is similar to Pivot Sampling, but incorporates two fundamental innovations: (i) the calculation of a perturbed **anchor pivot** that serves as a spatial reference, and (ii) the determination of a **calibrated adaptive radius** that restricts the perturbation domain to a region circular centered on the anchor.

3.4.1. Unified Scale Parameter Mapping

For the five levels of the unified scale (1 to 5), the same values of k are used as in Pivot Sampling:

Table 3.5: Mapping of privacy levels for ATP: values of k , calculated effective epsilon, and unified scale.

Level k	$\epsilon_{\text{effective}} = k \times N$	Interpretation
1	$1 \times 908 = 908$	Maximum privacy
2	$2 \times 908 = 1816$	High privacy
3	$3 \times 908 = 2724$	Moderate balance
4	$4 \times 908 = 3632$	Low privacy
5	$5 \times 908 = 4540$	Minimum privacy

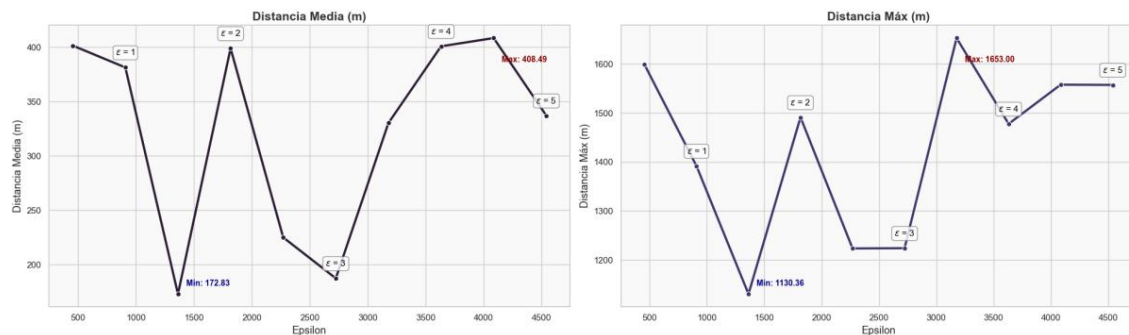
3.4.2. Evaluation of Metrics

The results of the ATP algorithm applied to the trajectory are presented below. real of Peking with $N = 908$ points, evaluated using the seven metrics described in the Chapter introduction. The values of $\epsilon_{\text{effective}}$ correspond to: level 1 ($\epsilon_{\text{eff}} = 908$), level 2 ($\epsilon_{\text{eff}} = 1816$), level 3 ($\epsilon_{\text{eff}} = 2724$), level 4 ($\epsilon_{\text{eff}} = 3632$), and level 5 ($\epsilon_{\text{eff}} = 4540$).

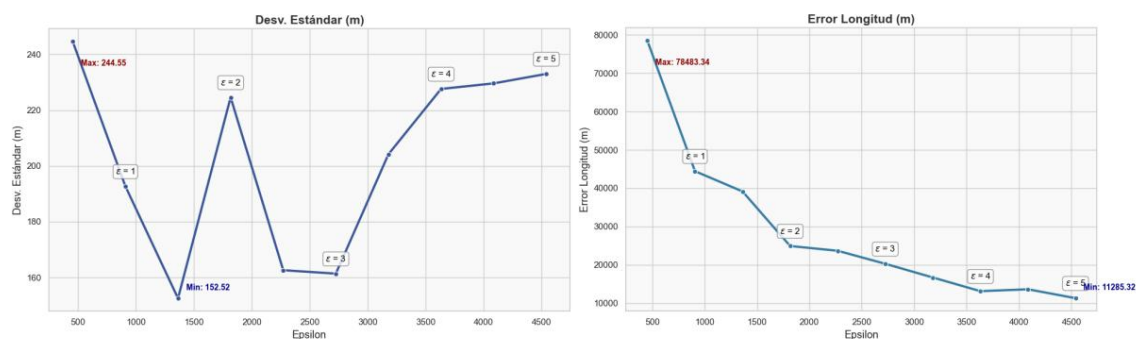
Average Distance. The average distance between original and perturbed points in ATP exhibits a non-monotonic behavior with a minimum at $\epsilon_{\text{eff}} = 1363$ (between levels 1 and 2), reaching 172.83 meters, and the maximum at $\epsilon_{\text{eff}} = 4086$ (level 4) with 408.49 meters. This pattern contrasts dramatically with the inversely proportional behavior observed in Laplace and Pivot Sampling, revealing a fundamental characteristic of ATP: the **nonlinear interaction between the privacy budget and the adaptive calibration of the radius**.

At level 1 ($\epsilon_{\text{eff}} = 908$), the average distance is 403.46 meters, higher than Pivot Sampling (253.50 m) and Laplace (241.78 m). This is due to the effect of the calibration term $\epsilon_{\text{eff}} \gamma$ which, with small γ , it significantly expands the adaptive perturbation radius R^* , increasing the domain P^* beyond the optimum. As γ increases towards level 2, the term $\epsilon_{\text{eff}} \gamma$ The exponential decays rapidly, allowing the adaptive constraint to operate effectively and reducing the average distance to its minimum of 172.83 meters.

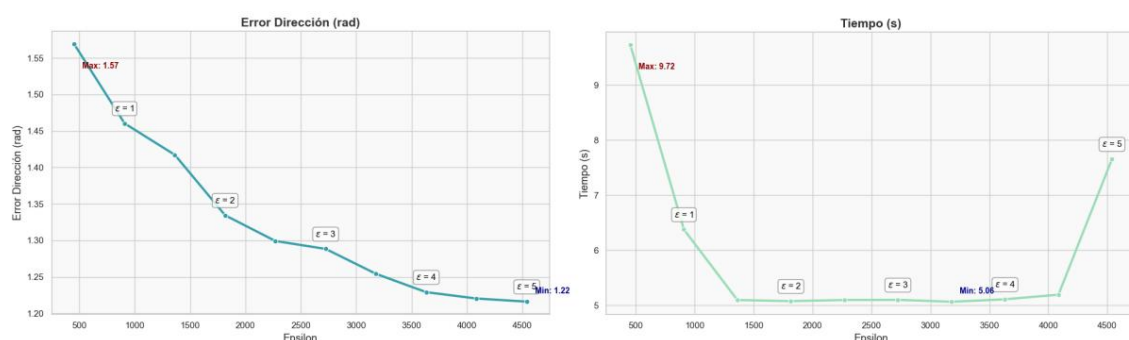
However, at intermediate levels (3-4), the average distance increases again, reaching values even higher than level 1. This counterintuitive behavior is explained by the convergence of the calibrated radius towards values that, although mathematically 'optimal according to' The formulation of ATP can result in P^* domains $\tilde{y}$ that include distant points when The geometry of the trajectory is irregular. At level 5 ($\tilde{y}_{\text{eff}} = 4540$), the distance decreases to 337.95 meters, although it remains significantly higher than Pivot Sampling (42.79 m) and Laplace (41.40 m).



(a) Average Distance (m) vs. Effective Privacy Budget $\tilde{y}_{\text{eff}} = k \times N$ for ATP. (b) Maximum Distance (m) vs. Effective Privacy Budget $\tilde{y}_{\text{eff}} = k \times N$ for ATP.



(c) Standard Deviation (m) vs. Effective Privacy Budget $\tilde{y}_{\text{eff}} = k \times N$ for ATP. (d) Length Error (m) vs. Effective Privacy Budget $\tilde{y}_{\text{eff}} = k \times N$ for ATP.



(e) Direction Error (rad) vs. Effective Privacy Budget $\tilde{y}_{\text{eff}} = k \times N$ for ATP. (f) Processing Time (s) vs. Effective Privacy Budget $\tilde{y}_{\text{eff}} = k \times N$ for ATP.

Figure 3.3: Error and performance metrics of the ATP algorithm under different values of the effective privacy budget $\tilde{y}_{\text{eff}}$.

Maximum Distance. The maximum distance in ATP exhibits an even more irregular pattern, oscillating between 1130.36 meters (minimum at $\gamma_{\text{eff}} = 1363$) and 1653.00 meters (maximum at $\gamma_{\text{eff}} = 3178$). At level 1 ($\gamma_{\text{eff}} = 908$), the maximum distance is 1598.43 meters, comparable to Pivot Sampling (1200.01 m) but greater than Laplace (1122.55 m). This metric reflects that, despite the domain restriction, ATP can generate extreme perturbations when the calibrated radius is large.

The behavior of the maximum distance reveals a significant limitation of ATP: the adaptive calibration of the radius, while effective in reducing the average distance in certain ranges, does not guarantee uniform control over extreme perturbations. At level 4 ($\gamma_{\text{eff}} = 3632$), where the maximum distance reaches 1459.20 meters, the mechanism appears to enter a regime where the expansion of the radius R^* due to the trajectory geometry dominates over the domain restriction, generating larger perturbations than at lower levels.

budget.

At level 5 ($\gamma_{\text{eff}} = 4540$), the maximum distance is 1568.91 meters, approximately 8.3 times greater than Pivot Sampling (187.84 m) and 8.2 times greater than Laplace (192.24 m) at the same level. This discrepancy indicates that ATP sacrifices control over extreme perturbations in favor of optimizing the average distance within specific ranges of γ .

Standard Deviation. The standard deviation of the displacement in ATP exhibits a pattern similar to the mean distance, with a minimum of 152.52 meters at $\gamma_{\text{eff}} = 1363$ and a maximum of 244.55 meters at $\gamma_{\text{eff}} = 454$ (very low level, prior to level 1). At level 1 ($\gamma_{\text{eff}} = 908$), the deviation is 193.50 meters, higher than Pivot Sampling (185.73 m) but lower than its own maximum, indicating that the algorithm is already in a phase of variability reduction.

The relationship between standard deviation and mean distance varies significantly across levels: approximately 0.48 at the minimum (level 1-2), increasing to 0.56 at level 4. This variation contrasts with the consistent relationship observed in Laplace (0.70) and Pivot Sampling (0.73), suggesting that ATP introduces heterogeneity into the perturbation pattern that is strongly dependent on the privacy budget.

At level 5 ($\gamma_{\text{eff}} = 4540$), the standard deviation is 233.16 meters, approximately 8.3 times higher than Pivot Sampling (28.13 m) and 8.0 times higher than Laplace (29.02 m).

This persistent discrepancy across all dispersion metrics indicates that ATP operates in a fundamentally different regime, where adaptive calibration amplifies variability rather than reducing it at high budget levels.

Length Error. The length error in ATP shows a more consistent decreasing trend than the point metrics, starting from 78,483.34 meters at $\gamma_{\text{eff}} = 454$ (prior to level 1) and progressively decreasing to 11,285.32 meters at $\gamma_{\text{eff}} = 4540$ (level 5).

At level 1 ($\gamma_{\text{eff}} = 908$), the error is 44,969.42 meters, significantly higher than Pivot Sampling (275,541.29 m) but of the same order of magnitude as Laplace (308,809.52 m).

Surprisingly, ATP achieves the lowest length error among the three algorithms at level 5: 11,285.32 meters compared to 25,508.38 meters in Pivot Sampling and 45,580.41

meters in Laplace. This represents a 56% improvement over Pivot Sampling and a 75% improvement over Laplace. suggests that, despite large point disturbances, the adaptive calibration of the radius It better preserves the overall path length structure in high budgets.

The smooth decreasing behavior of the length error contrasts with the fluctuations observed at medium and maximum distances, indicating that ATP achieves a favorable compromise in aggregate metrics even when point metrics are suboptimal. This is consistent with the hypothesis that domain restriction by spatial anchoring preserves local correlations between consecutive points, mitigating the quadratic accumulation of errors independent characteristic of Laplace.

Directional Error. The directional error in ATP decreases monotonically from 1.57 radians at $\gamma_{\text{eff}} = 454$ (prior to level 1) to 1.22 radians at $\gamma_{\text{eff}} = 4540$ (level 5). At level 1 ($\gamma_{\text{eff}} = 908$), the error is 1.47 radians, slightly lower than Laplace (1.86 rad) but higher to Pivot Sampling (1.83 rad). This trend is progressively reversed: at level 5, ATP It achieves 1.22 radians, lower than Laplace (1.68 rad) but still higher than Pivot Sampling (1.35 rad).

The monotonically decreasing behavior of the steering error, along with the error of length, constitutes evidence that ATP better preserves the **aggregated characteristics of shape** (global trajectory structure) than the **exact point positions**. The restriction of the domain through spatial anchoring introduces bias towards the central region of the trajectory, which favors the preservation of general directions even though the positions Individuals may deviate significantly.

The reduction in directional error is more moderate than in Pivot Sampling (1.57 to 1.22) rad, 22% reduction in ATP vs 1.83 to 1.35 rad, 26% reduction in Pivot Sampling), suggesting that the explicit directional perturbation of non-pivot points in Pivot Sampling It remains superior to the implicit spatial restriction of ATP to preserve orientation.

Processing Time. The processing time in ATP ranges from 5.06 seconds (minimum at multiple intermediate levels 2-4) and 9.72 seconds (maximum at $\gamma_{\text{eff}} = 454$, previous at level 1). At level 1 ($\gamma_{\text{eff}} = 908$), the time is 6.46 seconds, significantly more faster than Pivot Sampling (14.62 seconds) but considerably slower than Laplace (0.01 seconds).

The processing time in ATP is approximately 2.3-2.9 times shorter than Pivot Sampling at comparable levels, representing a substantial improvement in computational efficiency. This reduction is attributed to two factors: (i) the **restriction of the domain** $P^{\sim}$ reduce drastically the number of candidate points that must be evaluated during the disturbance, and (ii) the **simplification of the perturbation mechanism** by eliminating the two-stage structure complex Pivot Sampling (independent pivot disturbance + directional disturbance) (of no pivots with calculation of angular intersections).

The timing pattern shows an inverted U shape with maximum at the initial level and minimum at intermediate levels, then increasing slightly at level 5 (7.65 sec-

two). This behavior suggests that very low budgets require a greater number of iterations in radius calibration due to the large $e \cdot \ddot{y}$ term, while budgets

Very high values can expand the domain P^x increasing the number of evaluations.

Despite improvements over Pivot Sampling, ATP remains approximately 600 times slower than Laplace, which represents a significant limitation for real-time applications. The trade-off between computational efficiency and path structure preservation remains a fundamental challenge in spatially constrained LDP algorithms.

3.4.3. Comparative Analysis of Behavior

The Anchor-Based Pivot Sampling (ATP) algorithm introduces an adaptive approach that, while inspired by the Pivot Sampling mechanism, behaves markedly differently in response to variations in the privacy budget. Unlike traditional mechanisms, where increasing $\ddot{y}$ tends to consistently reduce the magnitude of the noise, ATP exhibits a non-monotonic pattern: the average distance reaches a minimum of 172.83 m at $\ddot{y}_{\text{eff}} = 1363$ and subsequently increases to 408.49 m at intermediate levels. This behavior is due to the complex interaction between the calibration term $\ddot{y}e\ddot{y}$ and the trajectory geometry, which causes the mechanism to operate optimally only within certain budget ranges, rather than maintaining progressive improvement with increasing budget.

of $\ddot{y}$.

ATP's main innovation lies in the adaptive restriction of the perturbation domain through spatial anchoring. Calculating the anchor pivot $p\ddot{y}$ and defining the domain $P \cdot \ddot{y} = p \cdot \ddot{y} \cdot P \mid \text{dist}(\hat{p}\ddot{y}, p) \cdot \ddot{y} R^{\wedge}$ allows perturbations to be concentrated in compact regions, following Tobler's First Law. This spatial restriction reduces the dispersion of perturbed points around the original path, although its effectiveness depends on the calibrated radius $R^{\wedge}$, which can expand excessively when $\ddot{y}$ is low or the path has irregular geometry.

The adaptive calibration process based on path geometry gives ATP significant flexibility. The parameter $\ddot{y}$, calculated using a weighted average of distances in the candidate set PR , automatically adjusts the algorithm's behavior according to the spatial structure of the data. This allows ATP to adapt to compact and uniform paths, improving its performance without the need for manual adjustments. However, this same geometric dependency introduces variability: dispersed paths or paths with multiple disconnected segments can lead to suboptimal results, as a single anchor fails to capture the complete path structure.

In terms of performance metrics, ATP shows a clear divergence between point and aggregate metrics. The former—such as mean distance, maximum distance, and standard deviation—exhibit high values and irregular behavior, while the aggregate metrics—such as length error and direction error—show sustained improvement. This pattern suggests that ATP prioritizes preserving the overall shape and consistency of the trajectory over point accuracy, sacrificing local accuracy in exchange for

maintain the overall structure of the movement.

Among the most relevant results, ATP achieves the lowest length error among the three algorithms compared, reaching 11,285.32 meters at level 5, which represents an improvement 56% compared to Pivot Sampling and 75% compared to Laplace. This performance stands out its ability to mitigate the accumulation of independent errors thanks to correlation spatial introduced by the anchoring, preserving the overall metric consistency of the route.

However, this advantage comes with notable limitations. In particular, ATP fails to adequately control extreme perturbations: the maximum distance oscillates between 1130.36 m and 1653.00 m, values considerably higher than those obtained by Laplace or Pivot Sampling. This occurs when the calibrated radius R^* is enlarged too much, including points distant from the anchor and generating excessive deviations. This phenomenon is accentuated in trajectories with high spatial dispersion, where the expansion of the P^* domain $\tilde{y}$ compromises the guarantees of local utility.

In terms of computational efficiency, ATP achieves an intermediate balance. With times With processing times between 5.06 and 9.72 seconds, the algorithm is between 2.3 and 2.9 times faster than Pivot Sampling, although it is still around 600 times slower than Laplace. The improvement in efficiency is mainly due to the reduction in the number of candidates within the restricted domain and the elimination of complex angular intersection calculations. characteristic of Pivot Sampling. The temporal pattern, shaped like an inverted "U", suggests that Very low and very high budget levels imply higher computational costs, while intermediate levels are more stable.

In summary, ATP positions itself as an adaptive local differential privacy mechanism that prioritizes the structural preservation of the trajectory through spatial restriction. Although it introduces notable advantages in the preservation of overall shape and relative efficiency, Its performance depends heavily on the geometric characteristics of the trajectory and it exhibits vulnerabilities to extreme perturbations. Therefore, ATP is particularly Suitable for urban mobility contexts where journeys follow regular spatial patterns, but less recommended for scenarios with dispersed or non-spatial trajectories structured.

Chapter 4

Experimental Analysis Graphs Origin-Destination

4.1. Description of the Experiment and Configuration Parameters cĩon

The experiment was designed to evaluate the comparative performance of three differential privacy mechanisms applied to trajectory data: the Laplace Mechanism, Pivot Sampling (TP), and Anchor-Based Pivot Sampling (ATP). For this purpose, the **Geolife GPS Trajectories** dataset developed by Microsoft Research [14, 15, 16] was used, which contains mobility records of 182 users in Beijing collected over three years. From this dataset, **484 real-world trajectories** located in adjacent areas within the Beijing urban area were selected to facilitate spatial comparison and reduce geographic bias during the analysis. Origin-Destination (OD) matrices were constructed from these trajectories, both for the original data and for the perturbed versions generated by each algorithm. These matrices allow us to quantify the spatial distortion introduced by privacy mechanisms in terms of flow between cells, serving as a basis for calculating the analyzed error metrics.

4.1.1. Privacy Budget Configuration

In this experiment, the same values of ϵ defined in **Chapter 3** were used, in order to maintain methodological consistency and allow direct comparisons between both analyses.

Similarly, the parameter k was again used for algorithms based on local differential privacy (LDP) to determine the effective value of ϵ as a function of the number of points N in each path. In contrast, for the Laplace Mechanism, which implements central differential privacy (CDP), the value of ϵ remains fixed and independent of N .

Table 5.2 summarizes the values used in this experiment, corresponding to the five

levels of the previously defined unified privacy scale (1 to 5).

Table 4.1: Privacy budget configuration used in this chapter.

the same values of $\tilde{y}$ defined in Chapter 3. For LDP-based algorithms, the effective budget is calculated as $\tilde{y}_{\text{effective}} = k \times N$, where N corresponds to the number of points in each path.

Laplace Mechanism	Level Pivot	Sampling / LDP Trace Interpretation	
1	$\tilde{y} = 0.01264$	$k = 1$ ($\tilde{y}_{\text{cash}} = N$)	Maximum privacy
	$\tilde{y} = 0.02148$	$k = 2$	High privacy
23	$\tilde{y} = 0.03032$	$(\tilde{y}_{\text{cash}} = 2N)$ $k = 3$ ($\tilde{y}_{\text{cash}} = 3N$)	Moderate balance
4	$\tilde{y} = 0.03916$	$k = 4$ ($\tilde{y}_{\text{cash}} = 4N$)	Low privacy
5	$\tilde{y} = 0.04358$	$k = 5$ ($\tilde{y}_{\text{cash}} = 5N$)	Minimal privacy

This configuration allows for direct comparability with the results presented in Chapter 3, ensuring that the variations observed in this analysis are due to solely to the experimental context and not to differences in the privacy parameters employed.

4.1.2. Step size variation and multiscale analysis

In addition to the level of privacy, the experiment incorporated a multiscale analysis to evaluate how **spatial resolution** affects the accuracy and stability of the derived metrics.

The study area was discretized using a regular geographic grid in which each cell represents a homogeneous spatial unit used to construct the Origin-Destination matrices (OD). The size of the side of each cell, defined by the step parameter m (in meters), controls the degree of spatial aggregation: smaller values generate a denser and more detailed mesh, while higher values produce a more aggregated and less sensitive representation to local noise. In this analysis, five levels of spatial resolution were considered:

Table 4.2: Step size values used in the experiment and their interpretation in spatial resolution terms.

Step size (m)	1000	Interpretation
	1500	High resolution / fine granularity
	2000	Moderately high resolution
	2500	Medium / balanced resolution
	3000	Low resolution / greater spatial aggregation
		Low resolution / large cells, distortion smoothing

corresponding to square cells with sides of 1 to 3 km. In this way, the following are covered scales that allow observation of both local mobility patterns (high resolution) and the most global trends of urban flow (low resolution).

Conceptual construction of the spatial grid. The general procedure for generating the The grid consists of projecting the area covered by the trajectories within a frame of

Geographical coordinates (latitude and longitude), which are then subdivided into regular intervals both in the north-south and east-west directions. Each point on the paths is assigned to a specific cell according to its position, which allows counting the displacements between cells of origin and destination and thus construct the corresponding OD matrix.

Unlike a predefined full grid, in this implementation the cells are **They are generated dynamically** as the trajectory coordinates are processed. Each When a coordinate appears for a cell that doesn't yet exist, a new entry is created in the cell map and assigned a unique identifier. This approach reduces the computational load and ensures that only areas actually traversed are represented, avoiding the inclusion of untraveled areas. presence of data.

To determine the extent of each cell in degrees, the step size (in meters) is converted using approximate scaling factors:

$$1_{\text{lat}}^{\circ} \approx 111,000 \text{ m}, \quad 1_{\text{lon}}^{\circ} \approx 111,000 \cos(\bar{y}) \text{ m}$$

where $\bar{y}$ is the mean latitude of the study area. This conversion allows defining the grid. directly in geographic coordinates without requiring additional cartographic projections, guaranteeing a consistent metric correspondence throughout the analyzed region.

Theoretical and empirical behavior. The total number of cells N_{cells} varies inversely with the square of the step size:

$$N_{\text{cells}} \propto \frac{1}{\text{step}^2}$$

This means that doubling the cell size reduces the number of spatial units to approximately one-quarter.

Empirically, this relationship holds approximately true: the exact number of cells depends on the spatial distribution of the trajectories (only effectively occupied cells are counted), on rounding effects in coordinate discretization, and on the small local variations in latitude that modify the longitudinal scale.

Overall, multiscale analysis allows us to identify the sensitivity of the metrics of error in relation to different degrees of spatial aggregation, providing a more complete view of the behavior of privacy mechanisms under different geographical resolutions.

4.2. Construction of the OD matrices and evaluation of results

Each of the mechanisms was applied independently to the 484 trajectories, generating distorted versions under different parameter configurations.

Subsequently, the perturbed positions were grouped into spatial cells to conform- sea the corresponding Origin-Destination (OD) matrices.

Based on the comparison between the original and perturbed matrices, the following were calculated Performance metrics: MAE, normalized MAE, relative error, and Frobenius norm

standardized.

These metrics allow us to quantify both local distortion (per cell) and coherence global structural mobility flow under each privacy configuration.

In summary, this experimental design allows for a systematic analysis of how the es-
The geometric structure of the trajectories and the formulation of the privacy budget—whether central or local—
influence the accuracy and stability of the perturbation mechanisms.
evaluated.

4.2.1. Analysis of Results of the Laplace Mechanism for Disturbance of Coordinates

The experimental results of the Laplace Mechanism are presented for different values of the privacy parameter ϵ and different spatial step sizes (1000 m, 1500 m, 2000 m, 2500 m, and 3000 m). The metrics used for the evaluation were the MAE, the normalized MAE, the relative error, and the normalized Frobenius norm, in order to quantify the impact of the disturbance on the fidelity of the Origin-Destination (OD) matrices.

In general, the behavior of the Laplace Mechanism confirms the inverse relationship between the privacy parameter ϵ and the magnitude of the error: as ϵ increases, the added Laplacian noise decreases, gradually reducing the distortion in the OD matrices.

The four metrics show decreasing trends, although with slight local fluctuations, particularly in the larger steps.

Consistency is maintained with the differential privacy principle: lower values of ϵ (greater noise) generate more pronounced deviations, while values above 0.03 lead to stabilized and consistent errors. Furthermore, step size has a moderating effect: steps of 1000–2000 m show greater sensitivity to variations in ϵ , while steps of 2500–3000 m exhibit flatter curves due to spatial aggregation, which smooths the errors.

Analysis of the MAE and normalized MAE The MAE reflects the mean absolute deviation between the original and perturbed OD matrices, while its normalized version allows comparison rare between different scales. Both indicators decrease steadily as it increases ϵ , confirming that Laplacian noise affects coordinates less when the privacy budget is higher.

At steps of 1000 and 1500 m, the MAE shows notable reductions (from approximately 1.4×10^{-2} to 3.5×10^{-3}), representing an improvement of more than 70% in accuracy. At larger steps, the mean error remains stable over wider ranges, but with increasing ϵ , indicating that spatial aggregation mitigates local noise variations. The normalized MAE reproduces this same trend with values on the order of 10^{-3} , showing a uniform decreasing pattern across configurations. , evidencing a

Relative Error Analysis The relative error was calculated cell by cell, dividing the absolute error by the original value of each OD flow, which provides a more accurate measure

50 4.2. CONSTRUCTION OF THE OD MATRICES AND EVALUATION OF RESULTS

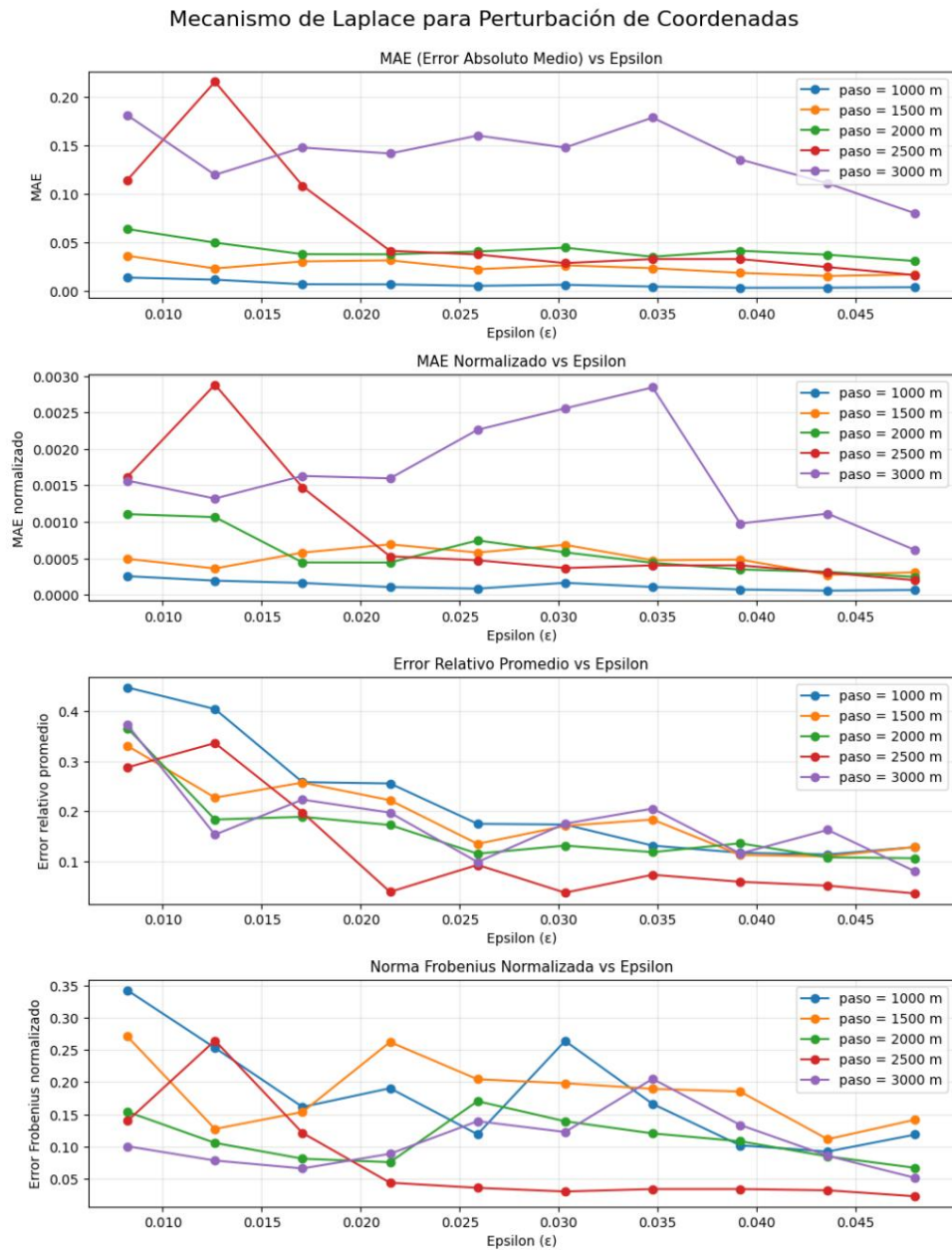


Figure 4.1: Results of the Laplace Algorithm for Coordinate Perturbation. The error metrics (MAE, normalized MAE, relative error and normalized Frobenius norm).

of proportionality. This metric shows a clear decrease with increasing $\tilde{y}$: from values close to 0.45 to approximately 0.11 for 1000 m steps, and from 0.33 to 0.11 in the case of 1500 m. This indicates that the disturbed flows retain their relative magnitude compared to the originals as the noise decreases.

In steps of 2500–3000 m, the relative error reaches minimums of around 0.05, although this The effect is attributed to spatial aggregation, which reduces local variations at the cost of losing granularity. Therefore, intermediate configurations ($\tilde{y} \in [0.03, 0.04]$, steps of 1500–2000 m) offer the most suitable balance between detail and stability.

Analysis of the normalized Frobenius norm The normalized Frobenius norm when Identifies the overall error across the entire OD matrix. The results show a decreasing pattern. sustained as $\tilde{y}$ increases, with values ranging from approximately 0.34 to 0.09 in steps of 1000 m. At intermediate steps (1500–2000 m), the metric stabilizes around 0.10–0.20 when $\tilde{y} \in [0.03, 0.04]$, indicating that the disturbances cease to introduce significant structural distortions into the flows. The decrease in the Frobenius norm reflects an improvement in the global spatial coherence of the OD system.

Comprehensive interpretation of the results The joint analysis shows that the Mechanism Laplace's function maintains predictable and stable behavior under coordinate perturbations. The decreasing pattern of the four metrics confirms the systematic reduction of error when increase the privacy budget, empirically validating the theoretical principles of the geometric differential privacy.

Furthermore, the comparison between different step sizes suggests that an intermediate granularity (1500–2000 μm) optimizes the relationship between accuracy and stability: steps too large Small steps amplify the local effect of noise, while large steps reduce spatial detail and excessively smooth flows.

In summary, the results confirm that the Laplace Mechanism is effective in preserving the privacy of coordinates without significantly compromising the usefulness of the data. The metrics show decreasing and stable behaviors for $\tilde{y} > 0.03$, with errors absolute and relative within acceptable margins. Therefore, it is concluded that configurations with $\tilde{y} \in [0.03, 0.04]$ and spatial steps between 1500 and 2000 m represent the equilibrium point Optimal balance between privacy and accuracy, guaranteeing differential protection and structural consistency of the OD flows.

4.2.2. Analysis of Results of the Pivot Sampling Algorithm for Privacy in Trajectories (TP)

This section analyzes the performance of the Pivot Sampling algorithm applied to the generation of Origin-Destination (OD) matrices under different privacy assumptions $\tilde{y}$ and different spatial granularities (step sizes). Each combination of parameters was evaluated using three error metrics: the normalized MAE, the relative error, and the norm normalized Frobenius, which quantify the degree of distortion introduced into the matrix

52 4.2. CONSTRUCTION OF THE OD MATRICES AND EVALUATION OF RESULTS

OD with respect to the original version. The experimental values correspond to the difference between the original OD matrices and the perturbed ones obtained after applying the privacy mechanism.

In general terms, the results confirm the performance of the Pivot algorithm. Sampling improves progressively as the privacy budget ϵ increases, consistent with differential privacy theory: the greater ϵ , the less injected noise and, therefore, the greater data fidelity. This behavior is visible in the sustained decrease of the three error metrics between $\epsilon = 1$ and $\epsilon = 5$, although with slight oscillations in some sizes of ϵ passed.

The step size (step m) also has a relevant effect on spatial stability of the model. Small steps (1000–1500 m) offer high spatial resolution but present greater variability in errors. In contrast, the intermediate steps (2000–2500 m) achieve a more stable balance between granularity and consistency, while large steps (3000 m) tend to lose structural precision and increase the overall error.

Analysis of the MAE and normalized MAE The MAE and its normalized version reflect the average magnitude of the absolute error between the original and disturbed OD flows, expressed respectively in absolute units and relative to total flows. Both metrics exhibit a general decreasing trend with increasing privacy budget ϵ , which confirms that a lower noise intensity leads to more accurate reconstructions of the mobility patterns.

In small steps (1000 m), the MAE decreases from 0.0167 for $\epsilon = 1$ to 0.0081 at $\epsilon = 4$, with a slight subsequent increase at $\epsilon = 5$. This behavior is replicated in the normalized MAE, which goes from 2.78×10^{-4} to 1.26×10^{-4} , indicating a reduction of approximately 55% in the average error. However, the trend is not strictly monotonous, since in some cases, the increase in the privacy budget produces slight fluctuations, attributable to the stochastic nature of pivoted sampling and the influence of cell size.

The global minimum value of both metrics is recorded for $\epsilon = 5$ and a step of 2000 m, with $MAE = 0.0408$ and $MAE_{norm} = 3.19 \times 10^{-4}$. This configuration shows that, with a high privacy budget and intermediate granularity, the algorithm manages to preserve from effectively the spatial coherence of OD flows.

Furthermore, at large steps (3000 m), both the MAE and its standardized version. They tend to increase again, reaching values of up to 0.29 and 2.10×10^{-3} respectively. This is due to the spatial over-aggregation effect, where the loss of detail. Directionality and compensation between adjacent flows degrade the accuracy of the re- matrix built.

In conclusion, the joint analysis of the MAE and the normalized MAE demonstrates that the precision of the algorithm consistently improves with increasing ϵ and that the intermediate steps (2000–2500 m) represent the optimal balance point between granularity and stability of the results.

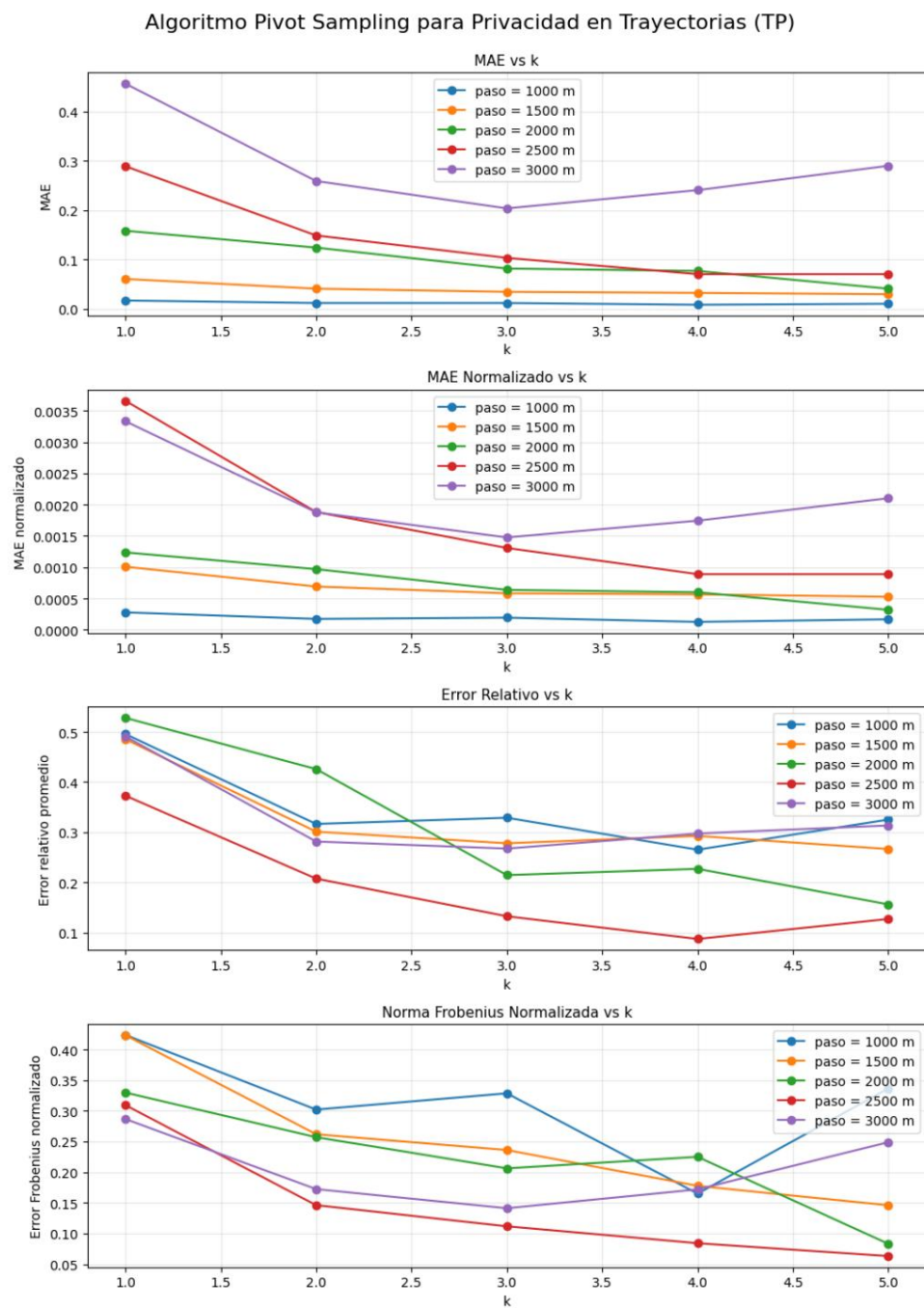


Figure 4.2: Results of the Pivot Sampling Algorithm for Path Privacy (TP). The three error metrics are illustrated (normalized MAE, relative error and Frobenius norm). standardized).

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Relative Error Analysis: The relative error evaluates the magnitude of the error with respect to the total original flows. The results show a significant reduction as $\tilde{y}$ increases, going from values close to 0.50 for $\tilde{y} = 1$ and minimums close to 0.09–0.13 for $\tilde{y} \in [4, 5]$. The smallest reported value ($Error_{rel} = 0.0877$) occurs at $\tilde{y} = 4$ and a step of 2500 m, which represents a reduction of the error of approximately 82

It is also observed that the intermediate steps (2000–2500 m) offer the best compromises between error and spatial stability, avoiding both excessive atomization of flows (in small steps) as the loss of directional resolution (in large steps).

Analysis of the normalized Frobenius norm The normalized Frobenius norm when It identifies the overall accumulated error over the entire structure of the OD matrix. Consistent with the previous metrics, a clear reduction is observed with increasing $\tilde{y}$. The minimum value is reached for $\tilde{y} = 5$ and a step of 2500 m, with a normalized Frobenius norm of 0.0631, which implies that the perturbed matrix retains more than 93

The sections between 2000 and 2500 m appear again as the most balanced, showing a stable and sustained decrease in the Frobenius norm without abrupt degradation in no range of $\tilde{y}$.

Comprehensive interpretation of the results Taken together, the three metrics confirm that The accuracy of the Pivot Sampling algorithm depends on the interaction between the budget of privacy $\tilde{y}$ and spatial granularity. For low values of $\tilde{y}$, angular and positional noise applied to pivots introduces significant distortions that affect spatial coherence of the trajectories. As $\tilde{y}$ increases, the mechanism preserves the trajectories more faithfully. direction and magnitude of displacements, reducing errors in the OD matrices.

Step size acts as a regulator of noise sensitivity: small steps maximize spatial resolution but amplify error variance, while steps Large values smooth the results at the cost of losing detail. Intermediate values (between 2000 and 2500 m) together with a high privacy budget ($\tilde{y} \in [4, 5]$) make up the configuration 'optimal, achieving overall error reductions of over 75

In summary, the Pivot Sampling algorithm maintains a solid performance against perturbation. local differential cion. The increase in the privacy budget $\tilde{y}$ consistently improves the precision of the Origin-Destination matrices, while an intermediate step size favors the stability of the model. Experimental results show that the strategy of Pivoted sampling manages to preserve the structural coherence of mobility flows even low moderate noise levels

4.2.3. Analysis of Results of the Anchor-Based Pivot Sampling Algorithm (ATP)

This section analyzes the experimental performance of the algorithm Anchor-Based Pivot Sampling (ATP), designed to preserve geometric differential privacy in trajectories through the incorporation of anchored pivots that reduce the directional dispersion of the

Disturbed points. Different configurations of the privacy budget $\tilde{y}$ were evaluated (represented by k) and the spatial step size (1000, 1500, 2000, 2500, and 3000 m). The metrics considered were the MAE, the normalized MAE, the relative error, and the norm of normalized Frobenius, in order to quantify the distortion introduced on the matrices Origin-Destination (OD) generated by the mechanism.

In general, the ATP algorithm exhibits more stable behavior than the Pivot algorithm. Traditional sampling, although with notable variations depending on the step size. Unlike the results obtained in the TP method, here a strictly monotonic decrease in errors is not observed when increasing $\tilde{y}$. The metrics tend to oscillate slightly.

especially in small and intermediate steps (1000–2500 m), suggesting an interaction complex between noise level, anchoring and spatial granularity.

In large steps (3000 m), however, the behavior stabilizes in a. Clearly: all error metrics decrease progressively as $\tilde{y}$ increases, demonstrating that the anchoring mechanism is more effective when the cells are wide and the trajectories. They exhibit less directional variability.

Analysis of the MAE and normalized MAE The MAE and its normalized version show a irregular trend in steps shorter than 2500 m. For example, for a 1000 m step, the values They remain around $MAE \tilde{y} 0.026$ and $MAE_{norm} \tilde{y} 4 \times 10^{-4}$, without a clear reduction with the increase of $\tilde{y}$. A similar situation occurs for steps of 1500 m and 2000 m, where The errors fluctuate between 0.09 and 0.21 in MAE, indicating that the mechanism introduces a magnitude of disturbance almost constant regardless of the privacy budget.

The best performance is observed in 3000 m steps, where the MAE gradually decreases from 0.498 for $\tilde{y} = 1$ to 0.367 for $\tilde{y} = 5$, and the normalized MAE goes from 3.4×10^{-3} to 2.5×10^{-3} . This suggests that anchoring improves its effect in scenarios of greater spatial aggregation, where The noise is distributed more homogeneously.

On the other hand, the intermediate steps (2500 m) present the greatest absolute errors ($MAE \tilde{y} 0.6$), possibly due to a combination of spatial detail loss and noise residual directional flux between nearby pivots, which generates a slight over-perturbation in the aggregate flows.

Relative error analysis The relative error presents a pattern similar to the MAE: high and oscillating in small and intermediate steps, with a downward trend only in large steps. For steps of 1000–2000 m, the values remain between 0.60 and 0.70, without a reduction significant as $\tilde{y}$ increases. In contrast, for a step of 3000 m, the relative error decreases from 0.63 at $\tilde{y} = 1$ to 0.47 at $\tilde{y} = 5$, representing an improvement of around 25%.

This indicates that, in more aggregated configurations, the algorithm is better able to preserve the proportions between OD flows, benefiting from the spatial averaging effect generated by the anchor.

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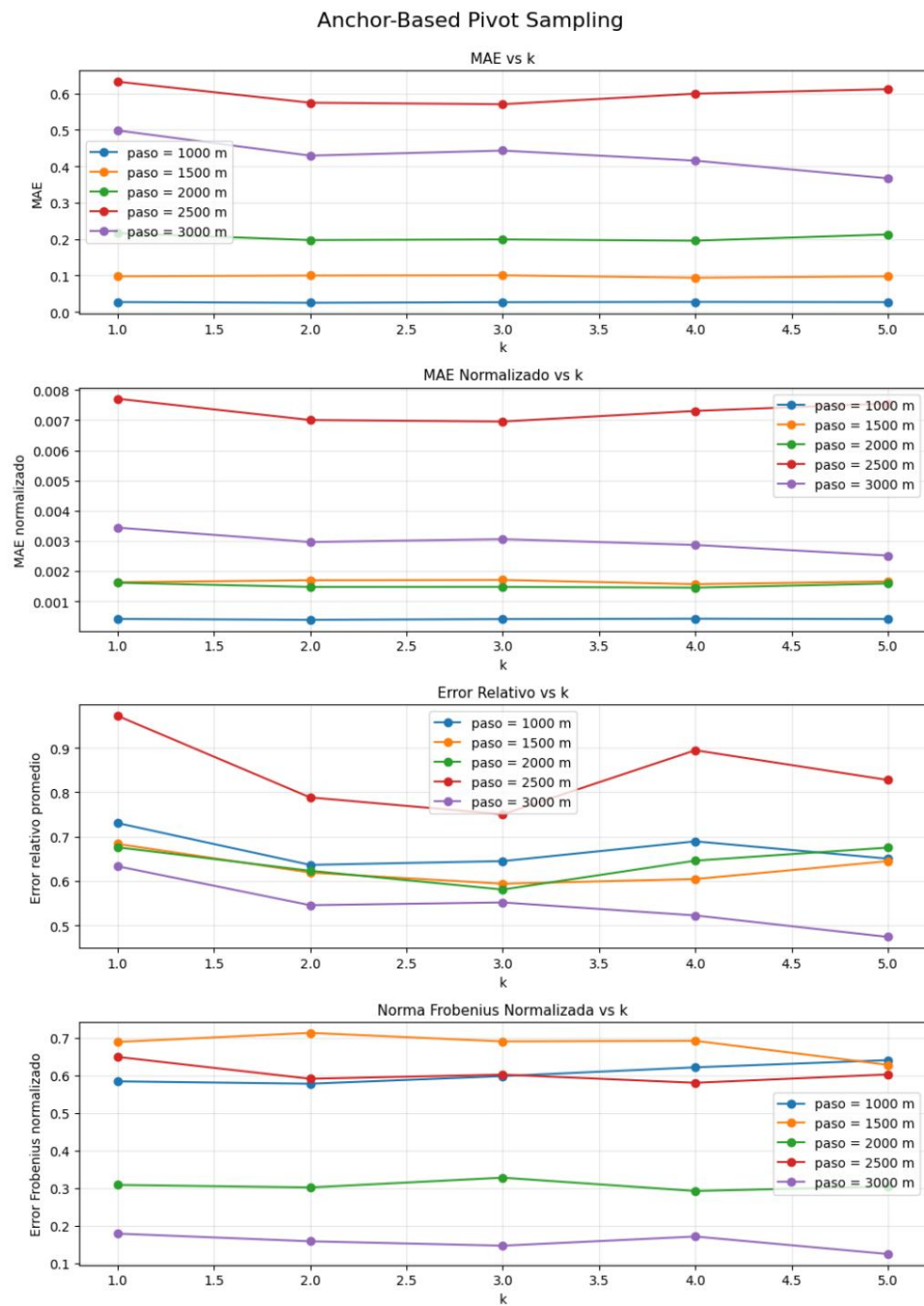


Figure 4.3: Results of the Anchor-Based Pivot Sampling (ATP) Algorithm. The error metrics (MAE, normalized MAE, relative error and normalized Frobenius norm).

Analysis of the normalized Frobenius norm The normalized Frobenius norm, which measures the overall error of the entire OD matrix, showing the most consistent pattern among the metrics. Although the initial values are relatively high for small steps ($\epsilon \sim 0.6$), the magnitude of the error is progressively reduced as the cell size increases, reaching its minimum globally in steps of 3000 m (0.125 for $\epsilon = 5$).

This demonstrates that, at large spatial scales, anchoring contributes to maintaining the global structure of flows, reducing accumulated dispersion and directional inconsistencies between disturbed trajectories.

Comprehensive interpretation of the results The set of results suggests that the ATP mechanism does not always reduce errors with increasing ϵ on fine scales, but it does improve consistently improves stability at larger scales. Anchoring introduces a spatial dependence between the pivots that attenuates local distortions, although this advantage is fully realized only when the cells are large enough to absorb the directional noise.

Therefore, ATP behaves as an effective mechanism in analytical contexts. cross-scope mobility, where the priority is to preserve the overall structural coherence of the trajectories rather than the short-range local details.

In summary, the experimental results show that the Anchor-Based Pivot Sampling (ATP) algorithm maintains a reasonable balance between privacy and utility, although its optimal performance is achieved for large pitch configurations (3000 m) and high values of ϵ ($\epsilon \sim 4$). In this regime, the three error metrics converge towards their minima, with reductions of up to 30% compared to the initial values.

The spatial anchoring of pivots allows for better preservation of the overall structure of the OD matrices, reducing accumulated dispersion without compromising the geometric consistency of the trajectories. Therefore, ATP emerges as a suitable method for scenarios where global structural stability is prioritized over local precision.

4.3. Global Comparison of Digital Privacy Mechanisms

referential

This section presents a comprehensive comparative analysis of the three mechanisms evaluated: the Laplace Mechanism for Coordinate Perturbation, the Pivot Sampling (TP) and Anchor-Based Pivot Sampling (ATP). The objective is to compare their behavior versus error metrics (MAE, normalized MAE, relative error and Frobenius norm) considering the influence of the privacy budget ϵ and the spatial step size.

The overall analysis reveals three distinct performance patterns. The Laplace Mechanism exhibits a clear and regular decreasing trend in all metrics as ϵ increases, with predictable and stable behavior. This is because Laplacian noise is applied from uniformly and independently over each coordinate, generating local disturbances

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bounded and without spatial correlation. Consequently, their errors decrease monotonically and controlled, reaching stable minimum values for $\tilde{\gamma} \approx 0.03$.

In contrast, the Pivot Sampling algorithm introduces greater variability in the results due to the directional nature of the noise applied to the pivots. Although it follows the same general trend of error reduction with increasing $\tilde{\gamma}$, fluctuations are more noticeable in small steps. Pivoted sampling generates displacements dependent on the geometric context of the trajectories, which increases sensitivity to spatial granularity. However, in configurations with intermediate steps (2000–2500 m) and high values of $\tilde{\gamma}$, the algorithm achieves a remarkable balance between precision and structural coherence, with reductions of global error of up to 80%.

Finally, the Anchor-Based Pivot Sampling (ATP) mechanism shows a pattern of more complex behavior. The use of anchored pivots introduces spatial correlations, which partially stabilize the directional distortion, but the magnitude of the error does not decrease in a strictly monotonic manner with $\tilde{\gamma}$. At fine scales (1000–2000 m), the anchorage generates local oscillations due to the interaction between directional noise and spatial dependence; however, in large steps (3000 m), the algorithm stabilizes and surpasses in global consistency the TP, with error reductions close to 30%.

Comparison of the MAE and normalized MAE In terms of mean error (MAE) and its normalized version, the Laplace Mechanism maintains the lowest absolute values for high privacy settings (low $\tilde{\gamma}$), which positions it as the most stable method in noisy environments. Pivot Sampling progressively improves its accuracy as $\tilde{\gamma}$ increases, reaching its best performance in intermediate steps, where the relationship between spatial resolution and angular noise is more pronounced. ATP, on the other hand, exhibits an effective reduction of the MAE only in large steps, indicating that its anchoring mechanism is more efficient at the macroscopic level than at local scales.

In summary, the Laplace transform excels in local accuracy, while the TP transform excels in stability at medium resolution, and ATP in structural coherence when spatial aggregation is greater.

Comparison of relative error. The relative error reinforces these conclusions. The Laplace Mechanism uniformly reduces the error proportion with increasing $\tilde{\gamma}$, reaching minimums close to 0.05 in aggregated configurations. Pivot Sampling achieves values similar to Laplace's (around 0.08–0.10) when $\tilde{\gamma} \approx 4$ and the step size is between 2000 and 2500 μm , reflecting good conservation of OD flux ratios. ATP, on the other hand, maintains higher relative errors in fine steps (between 0.60 and 0.70), but improves significantly in 3000 m steps, where the relative error drops to 0.47; this suggests that the Anchoring redistributes noise more homogeneously in lower resolution contexts.

Comparison of the normalized Frobenius norm The normalized Frobenius norm allows for the evaluation of the overall preservation of the structure of the OD matrices. The Laplace transform shows the most uniform and sustained descent, with minimums close to 0.09 in steps of 1500–2000 m.

The Pivot Sampling achieves comparable performance, reaching minimum values of 0.06–0.07 in optimal configurations of $\gamma = 5$ and intermediate step. For its part, ATP reduces its Frobenius standard of 0.58 to 0.12 in 3000 m steps, consolidating itself as the method more robust in terms of overall structural preservation, although not necessarily the most locally accurate.

Comprehensive interpretation and comparative synthesis From a comprehensive perspective, the three mechanisms complement each other in their range of application:

- **Laplace:** optimal for perturbation of individual coordinates or high spatial resolution studies, where a direct and controlled relationship between privacy and precision.
- **Pivot Sampling (TP):** suitable for trajectories with medium resolution, where the direction of movement is relevant and the aim is to preserve local coherence without over-aggregation the data.
- **Anchor-Based Pivot Sampling (ATP):** more efficient in macroscopic or aggregate analyses, where the priority is to maintain the overall flow structure and minimize dispersion. cumulative torsion between zones.

In conclusion, the comparison reveals a gradient of specialization between the mechanisms:

The Laplace offers local precision under high privacy, the Pivot Sampling balances privacy and utility in intermediate contexts, and ATP maximizes overall structural stability when

It prioritizes spatial coherence over individual detail. Therefore, the choice of mechanism

The most appropriate one will depend on the level of spatial aggregation and the type of mobility analysis that one wishes to preserve under the framework of geometric differential privacy.

Chapter 5

Graphical Analysis of Trajectories

This chapter presents the tool developed for the interactive visualization of GPS trajectories and the graphical evaluation of the effect of differential privacy mechanisms. The tool is publicly available at: https://cflorido.github.io/front_thesis/.

The application allows the user to:

- Select the differential privacy **algorithm** to analyze: Laplace mechanism, Pivot Sampling (TP) or Anchor-Based Pivot Sampling (ATP).
- Choose the desired **type of display** , from several options detailed below. forward.
- Adjust the **experiment parameters**: the step size (step m) and the budget of privacy (ϵ).
- Alternate the display between the **original** data and the **disturbed** data .

5.1. Configuration Parameters

To maintain consistency with previous experiments, the same values are used for step size and privacy budget defined in Chapter 3.

Table 5.1: Step size values used in the experiment and their interpretation in spatial resolution terms.

Step size (m)	Interpretation
1000	High resolution / fine granularity
1500	Moderately high resolution
2000	Medium / balanced resolution
2500	Low resolution / greater spatial aggregation
3000	Low resolution / large cells, distortion smoothing

Table 5.2: Privacy budget configuration used in this chapter. For the In LDP-based algorithms, the effective budget is calculated as $\epsilon_{\text{effective}} = k \times N$, where N corresponds to the number of points in each path.

Laplace	Mechanism	Level Pivot Sampling / LDP	Trace Interpretation
1	$\epsilon = 0.01264$	$k = 1$ ($\epsilon_{\text{cash}} = N$)	$k = 2$
2	$\epsilon = 0.02148$	$(\epsilon_{\text{cash}} = 2N)$	$k = 3$ ($\epsilon_{\text{cash}} = 3N$)
3	$\epsilon = 0.03032$	$k = 4$ ($\epsilon_{\text{cash}} = 4N$)	
4	$\epsilon = 0.03916$	$k = 5$ ($\epsilon_{\text{cash}} = 5N$)	
5	$\epsilon = 0.04358$		
			Maximum privacy High privacy Moderate balance Low privacy Minimal privacy

5.2. Types of Visualization

The tool developed allows exploration of trajectory data and its perturbations through five types of visualization, each with a different focus on the structure spatial and mobility flows:

1. **Origin-Destination (OD) Matrix:** This visualization represents the number of paths that move from a source cell to a destination cell. Each element (i, j) The matrix indicates the flow of paths between corresponding cells. It allows Identify frequent routes and aggregate mobility patterns. Comparing the matrix The original, along with the disturbed one, allows us to evaluate how privacy mechanisms affect the preservation of flows.



Figure 5.1: Example of Origin-Destination matrix generated by the tool.

2. **Destination Cell Heat Map:** Each cell is colored according to the number of journeys that end there, generating an arrival density map. Cells with Colors closer to yellow represent frequent destinations and allow visualization Concentration of activity in the city. This visualization helps to identify areas destination critiques and evaluate how data perturbation can alter perception of spatial concentration.

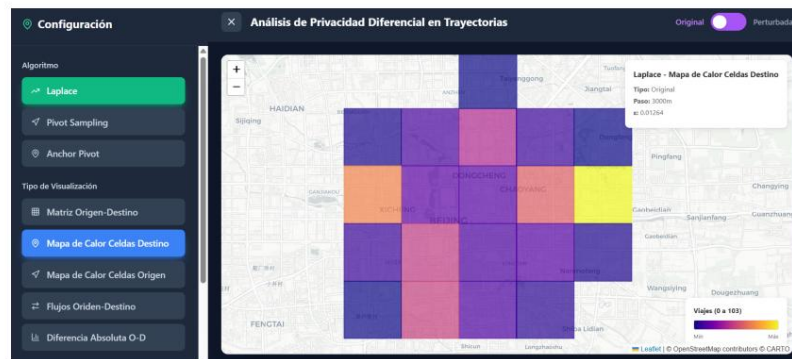


Figure 5.2: Heat map of target cells.

3. **Source Cell Heat Map:** Similar to the previous one, but considering the starting point of the trajectories. It allows identification of the main starting points of the movements and how privacy mechanisms affect the visibility of the patterns.
- to begin with.

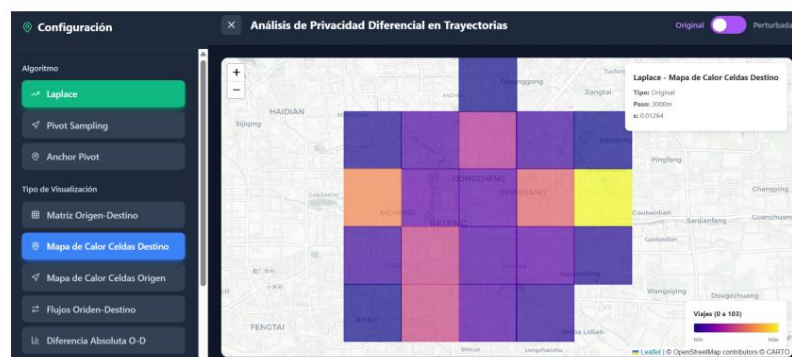


Figure 5.3: Source cell heat map.

4. **Combined Origin-Destination Flow Map:** This map combines a heat map of origin cells (blue scale) and destination cells (red scale), along with flow lines connecting the start and end cells of each path. It provides a comprehensive view of the spatial concentration and direction of movements, which is critical for understanding both the intensity and direction of displacements and how privacy mechanisms modify the flow structure.



Figure 5.4: Combined Origin-Destination flow map with overlapping source and destination cells destination.

5. **Absolute difference per cell between original and perturbed OD matrix:** Each cell This represents the difference in the number of paths between the original and disturbed data. The more yellow colors indicate greater distortion in that cell, allowing To spatially locate where privacy mechanisms introduce the greatest changes. This visualization is essential for evaluating the impact of differential privacy. at a granular level and to identify specific distortion patterns induced by each algorithm.



Figure 5.5: Map of absolute difference per cell between original and perturbed OD matrix.

5.3. Interaction with the Tool

The interface allows the user to:

- Alternate between the visualizations mentioned to observe how the patterns change spatial according to the type of analysis.

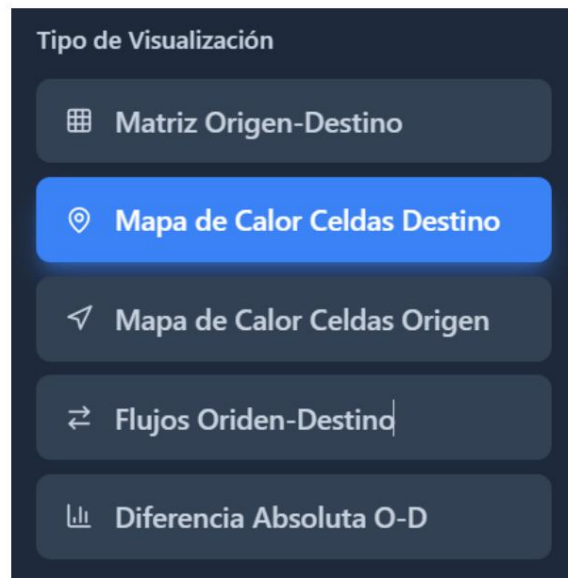


Figure 5.6: Button to alternate visualizations.

- Modify the step size (step m) and the privacy budget (ϵ) to study the sensitivity of the results to these parameters.

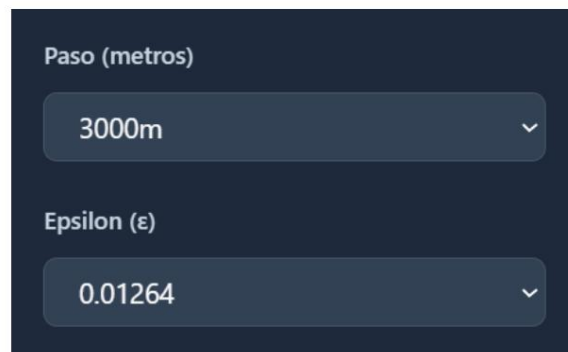


Figure 5.7: Panel for changing display and privacy parameters.

- Comparing the original data against the disturbed data allows for a visual assessment of the impact of each differential privacy mechanism.



Figure 5.8: Interface for switching between original and disturbed data.

- Select the privacy algorithm you wish to analyze.

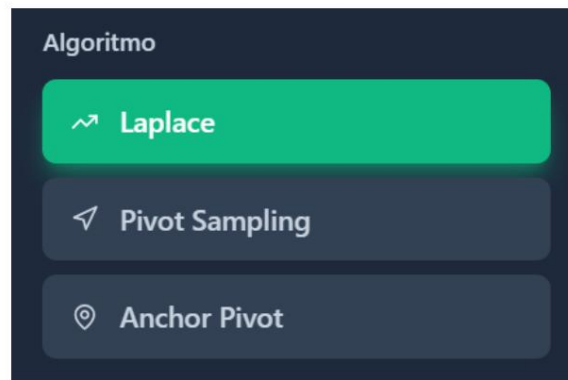


Figure 5.9: Algorithm selector in the interface.

- Click on each cell in all visualizations to obtain additional information, such as the cell value, cell ID, or corresponding trips.



Figure 5.10: Additional information when clicking on a cell.

5.4. Final considerations

This tool facilitates multiscale exploration of trajectories and graphical evaluation of differential privacy mechanisms. The combination of algorithm selection, adjustment of parameters and different types of visualization allows for a complete analysis of the distortion. The range of parameters and different types of visualization allows for a complete analysis of the distortion. spatial and mobility patterns in GPS data.

Chapter 6

Conclusions and Future Work

This paper addresses the problem of privacy protection in location data through the application of Differential Privacy (DP) mechanisms. Throughout the study, a comparative evaluation was developed between different perturbation strategies—including the Laplace Mechanism, Pivot Sampling (PS), and Anchor-Based Pivot Sampling (APS)—in order to analyze their impact on the usefulness of the data and their ability to preserve meaningful spatial patterns.

The experimental framework included both quantitative metrics (mean distance, standard deviation, angular error, among others) and interactive visual representations that facilitated the understanding of the effects of each mechanism on the GPS trajectories. This combination allowed for a comprehensive view of the trade-off between privacy and utility, offering a robust approach for the analysis of spatial data under anonymized conditions.

6.1. Discussion of Results

The results obtained show that the Laplace Mechanism, although conceptually simple and widely used, introduces considerable distortion in trajectories with fine spatial structures, especially when the privacy budget ϵ is low. In contrast, pivot point (TP) and ATP-based mechanisms achieve better conservation of motion topology, maintaining spatial coherence and trajectory continuity even under stricter privacy levels.

In particular, the Anchor-Based Pivot Sampling (ATP) mechanism demonstrated greater stability with increasing point numbers, making it more suitable for scenarios with dense or long-duration trajectories. This behavior suggests that spatial anchoring techniques can offer a better balance between structural preservation and privacy protection.

Visual analysis complemented the numerical results, revealing how the shape and distribution of trajectories change as the privacy budget decreases.

The distortion manifests itself unevenly in space, being more pronounced in areas with higher movement density. These findings reinforce the need for adaptation

the mechanisms of perturbation to the spatial context of the data.

6.2. Visualization Tool

The visualization tool developed constitutes a practical and methodological contribution. relevant logic. It allows interactive exploration of the relationship between the level of privacy and the spatial distortion introduced, offering visual support for the interpretation of experimental results.

Its modular and multi-platform design facilitates the integration of new mechanisms privacy and its extension to different types of geographic data. Consequently, the This tool can be used in both research and educational settings. to intuitively illustrate the principles of differential privacy and its impact on the trajectory analysis.

6.3. Potential Applications

The results and tools developed in this thesis offer broad potential applicable in areas where geographic information is an essential input, but where the protection of individual privacy is equally a priority. Among the most important areas Relevant aspects include urban mobility, vehicular traffic management, and planning. public transport, the assessment of the environmental impact of travel and studies spatial behavior of the population.

The source-destination (OD) matrices generated from the perturbed data represent a particularly valuable tool. They allow for the estimation of aggregate mobility flows between geographical areas, without needing to access individual trajectories, which is crucial For public and private entities that need to analyze transportation demand, design optimal routes, or evaluate the efficiency of existing infrastructure. Thanks to the application of local differential privacy mechanisms, these matrices retain statistical utility necessary for decision-making, while significantly reducing the risk of user re-identification.

Similarly, the visual maps built in the developed tool allow to intuitively observe the spatial distribution of displacements and alterations introduced by the various disturbance mechanisms. These visualizations can be used by researchers and urban planners to understand the behavior of The mechanisms, under different privacy parameters, can identify aggregated mobility patterns or communicate results in a way that is understandable to non-technical audiences. The possibility Comparing original and perturbed trajectories also facilitates the qualitative evaluation of balance between privacy and utility.

In a broader context, the methods and visualizations implemented in this thesis They can be integrated into geospatial analysis platforms that require sharing sensitive data. location-based—for example, in smart city applications, flow monitoring

tourist or territorial accessibility studies—, complying with data protection regulations and ethical principles for handling personal information. In summary, this work not only demonstrates the technical feasibility of applying local differential privacy to mobility data, but also paves the way for practical tools for the responsible and secure use of spatial information in research and public administration.

6.4. Limitations of the Study

Although the experiments were conducted on actual user journeys, the pre- This presents limitations that must be taken into account when interpreting the results. First, although the data are real, the analysis focused on a representative subset and specific urban contexts, which may introduce geographical and behavioral biases that limit generalization to other cities, rural contexts, or different mobility patterns. Second, although individual trajectories are authentic, the study did not comprehensively model the spatial and temporal interdependencies between multiple users—for example, spatiotemporal coincidences between trajectories or crowd effects—factors that in real-world deployments can influence the risk of re-identification and the effectiveness of perturbation mechanisms. Third, the temporal dimension was treated in a limited way: variations in sampling frequency, seasonality, and complex temporal dynamics that could alter both the usefulness and privacy guarantees in operational applications were not explored in detail. Finally, there are practical limitations related to the selection and calibration of parameters (choice of γ , adaptive radii, spatial granularity) and to the computational scalability of the more sophisticated methods. These limitations affect the feasibility of deploying some variants of the algorithm in environments with large volumes of data or strict latency requirements.

Likewise, the analysis focused on the spatial dimension, leaving open the possibility of incorporating the temporal dimension to evaluate the impact of privacy in time series or trajectories with high sampling frequency.

6.5. Future Work

As future work, it is proposed to extend the study to dynamic contexts where the level of privacy varies adaptively depending on the sensitivity of the environment or the type of user. It would also be relevant to evaluate the behavior of machine learning models trained on perturbed data, analyzing how the different LDP mechanisms affect predictive capacity or the quality of spatial clustering.

Another relevant line of research involves exploring the combination of probabilistic approaches with spatial representations based on graphs or geographic neural networks. This would allow the construction of hybrid models that preserve global patterns while maintaining individual privacy.

Finally, it is considered valuable to continue developing the visualization tool.

to incorporate advanced analytical functionalities, such as automatic comparison between mechanisms, generation of spatial fidelity metrics, or simulation of custom privacy scenarios.

6.6. Final Conclusion

In conclusion, this thesis contributes to the understanding of the impact of local differential privacy on location data, offering both an empirical analysis of its behavior and a practical tool for visualization and interpretation. The balance between privacy and utility is consolidated as a fundamental challenge in the era of ubiquitous data, and the mechanisms evaluated in this work represent a significant step towards more secure, transparent, and ethically responsible solutions.

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